

BFS Traversal

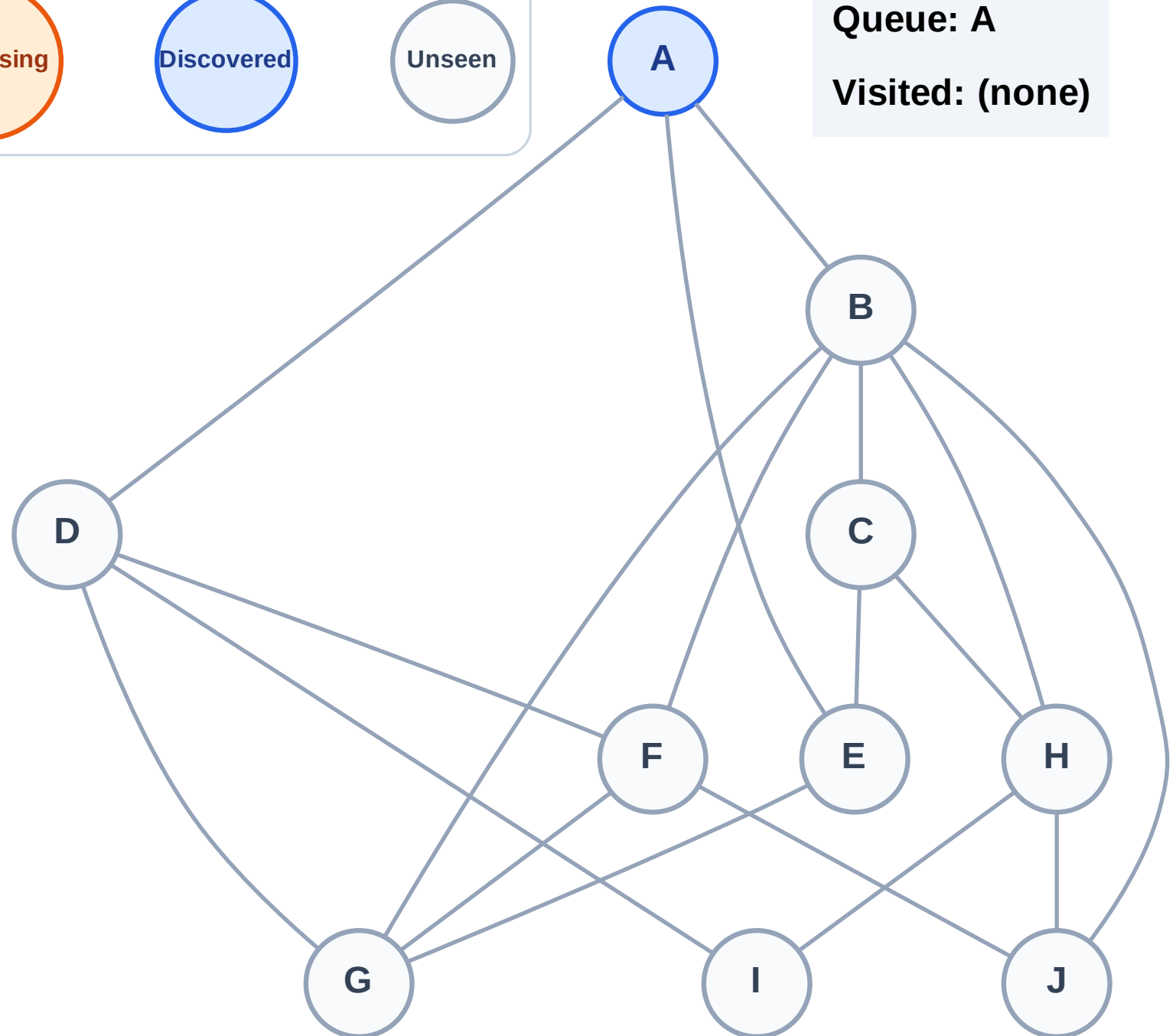
Step 1: Start at A

Legend



Queue: A

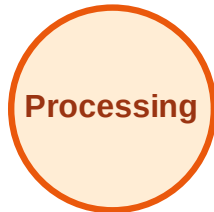
Visited: (none)



BFS Traversal

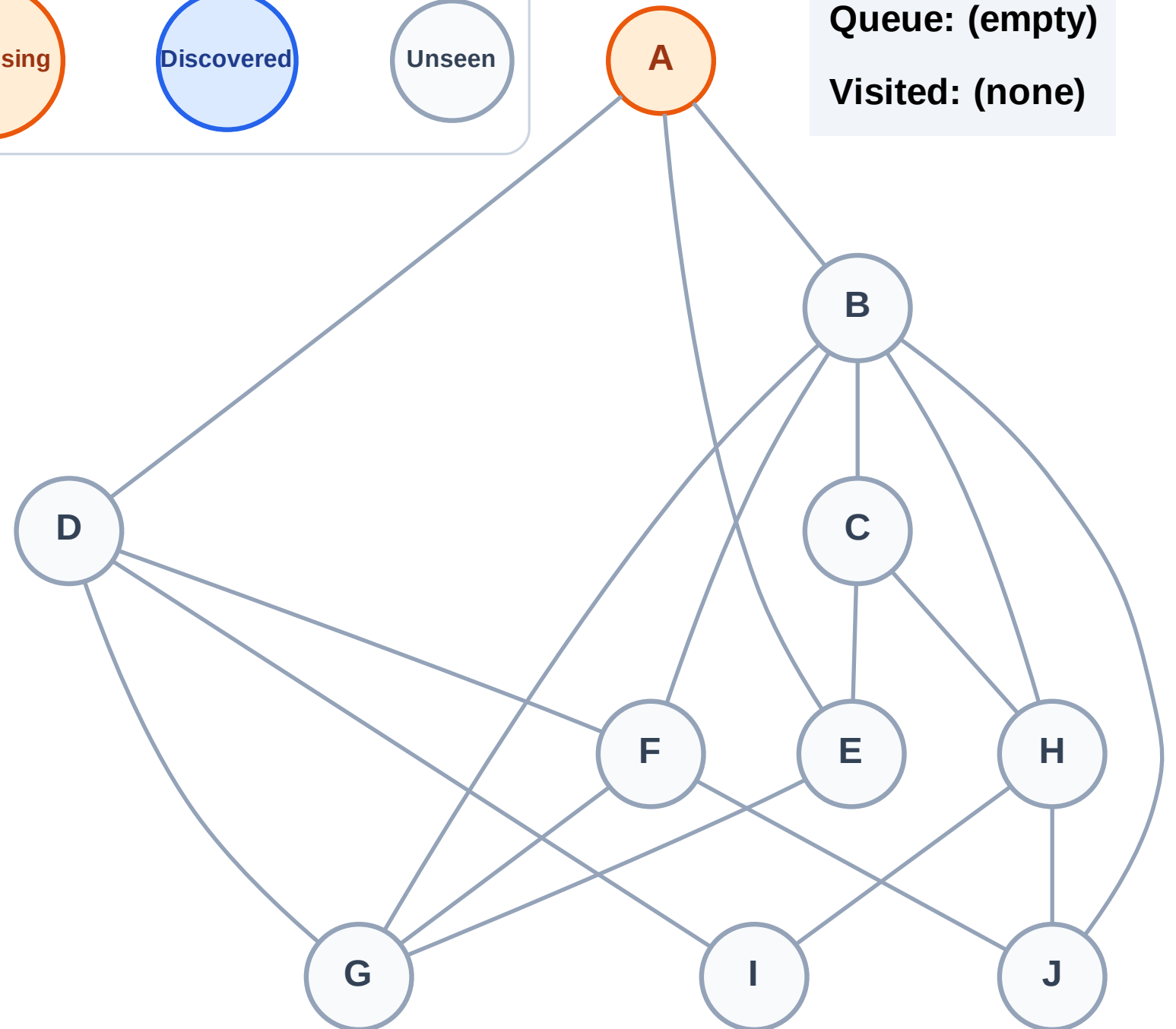
Step 2: Process node A

Legend



Queue: (empty)

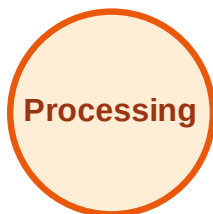
Visited: (none)



BFS Traversal

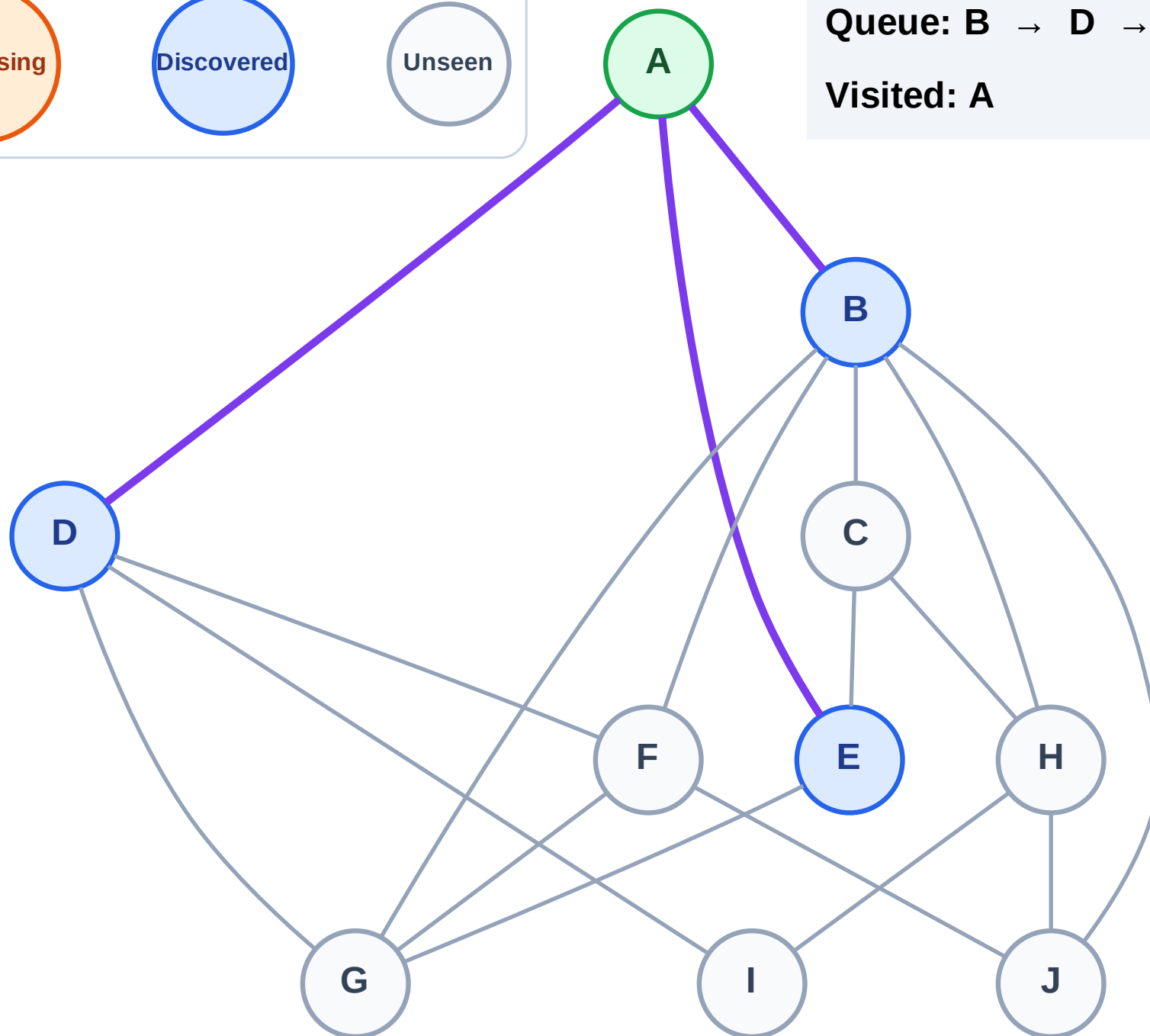
Step 3: Discover B, D, E from A

Legend



Queue: B → D → E

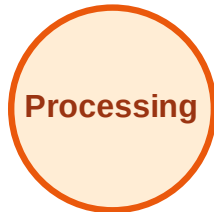
Visited: A



BFS Traversal

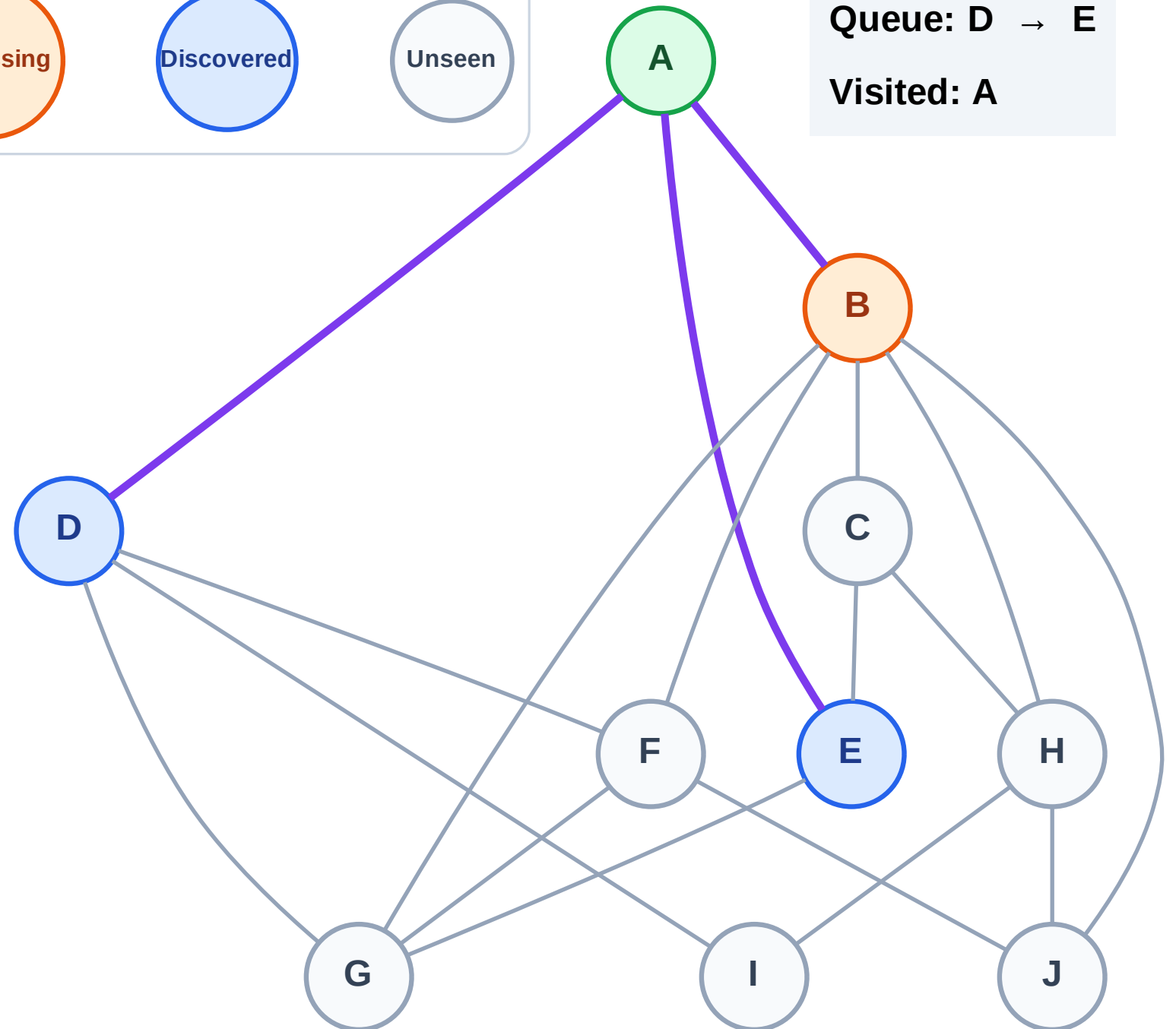
Step 4: Process node B

Legend



Queue: D → E

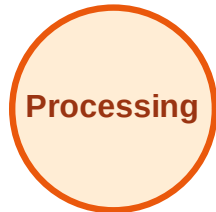
Visited: A



BFS Traversal

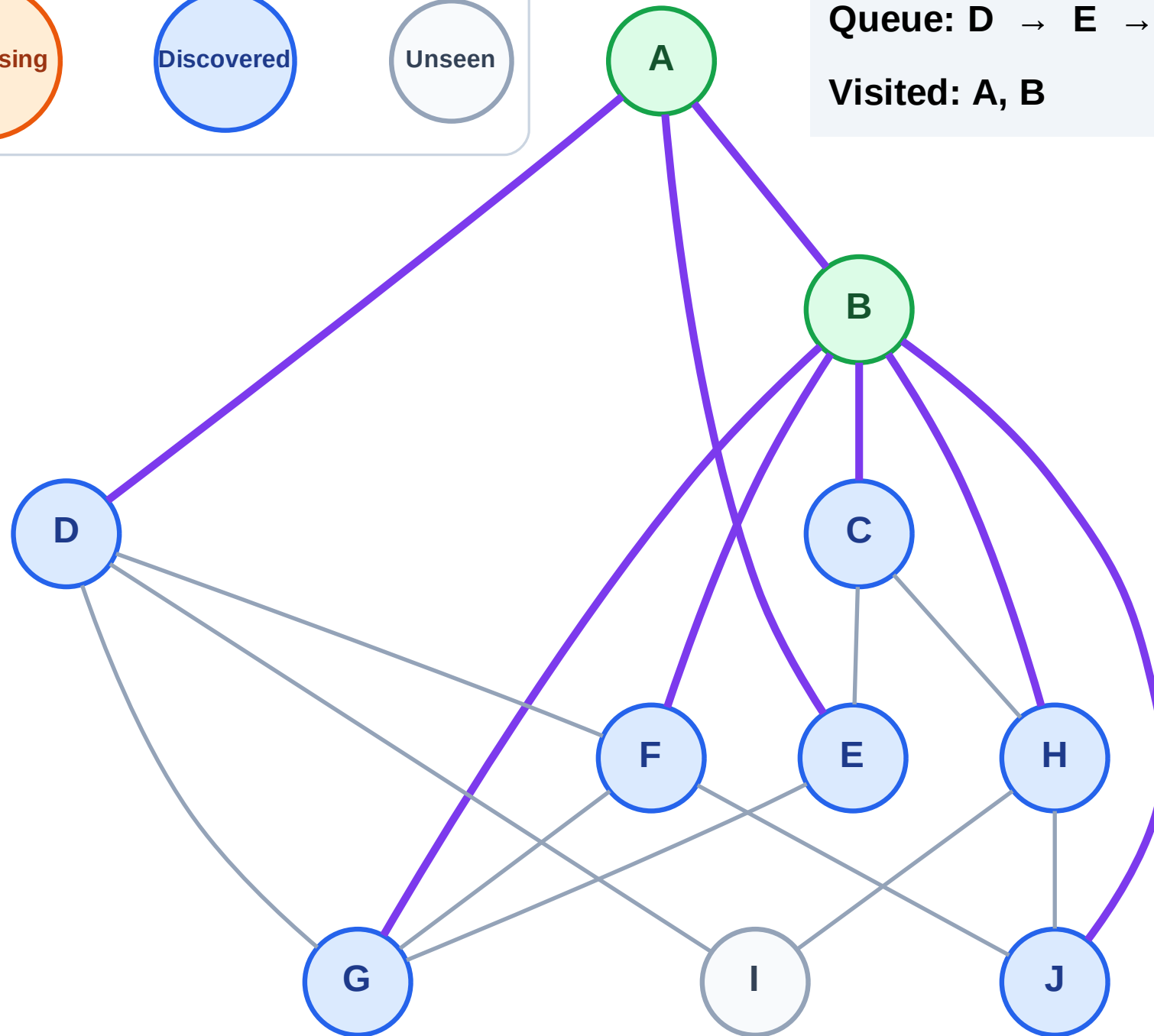
Step 5: Discover C, F, G, H, J from B

Legend



Queue: D → E → C → F → G → H → J

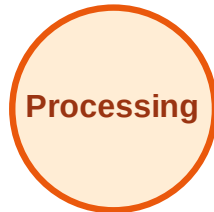
Visited: A, B



BFS Traversal

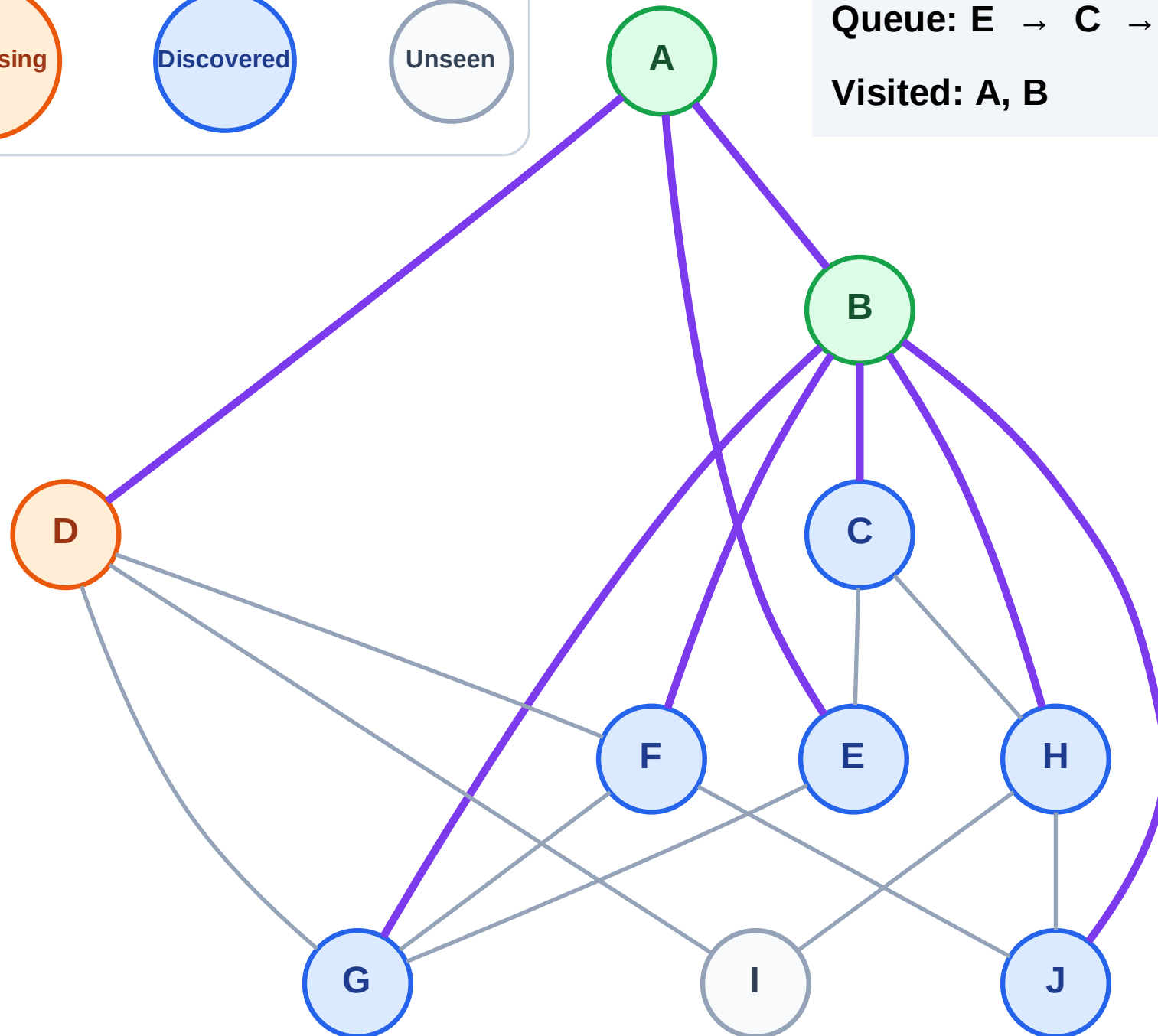
Step 6: Process node D

Legend



Queue: E → C → F → G → H → J

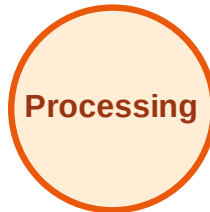
Visited: A, B



BFS Traversal

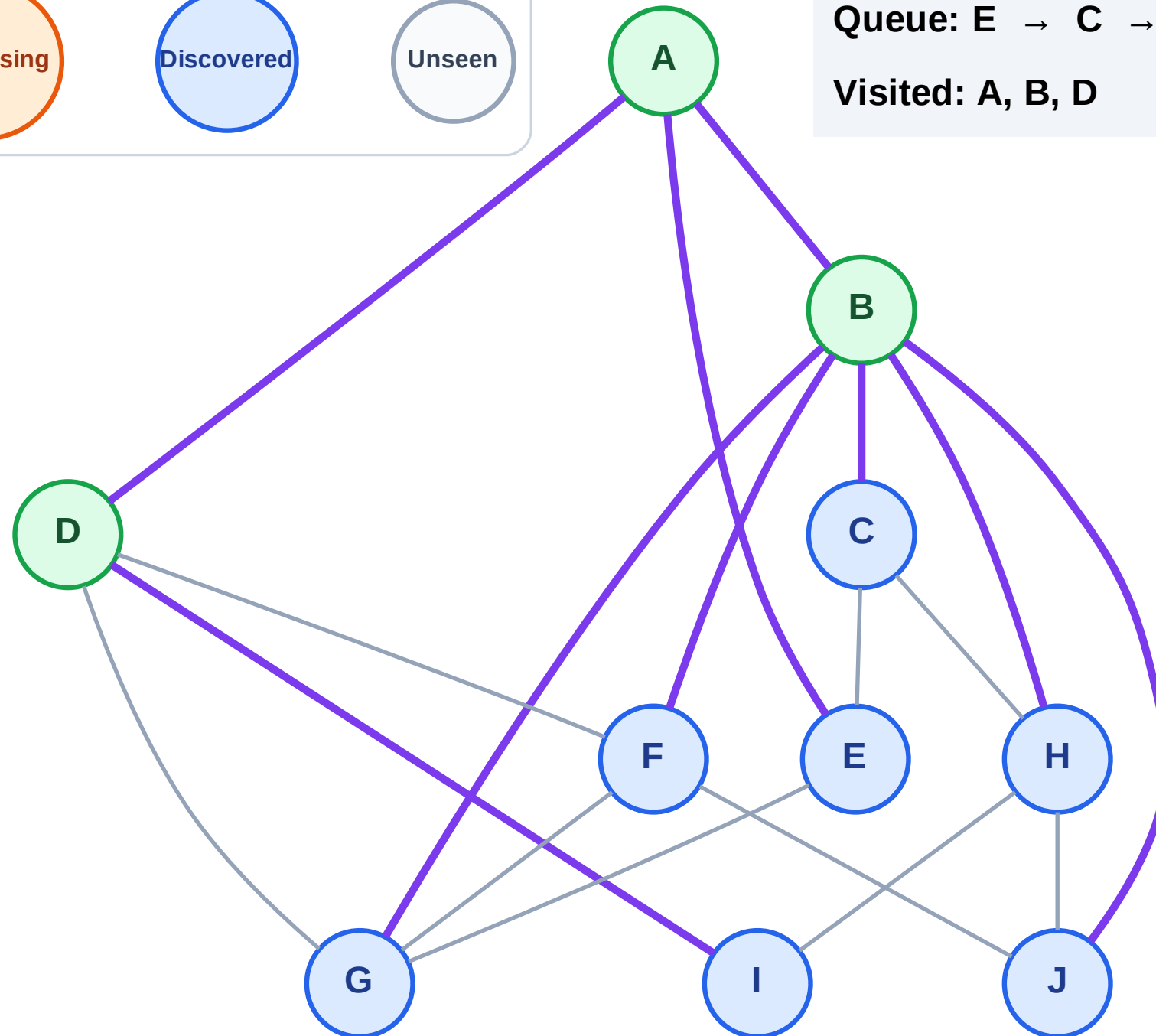
Step 7: Discover I from D

Legend



Queue: E → C → F → G → H → J → I

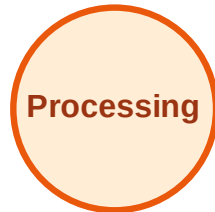
Visited: A, B, D



BFS Traversal

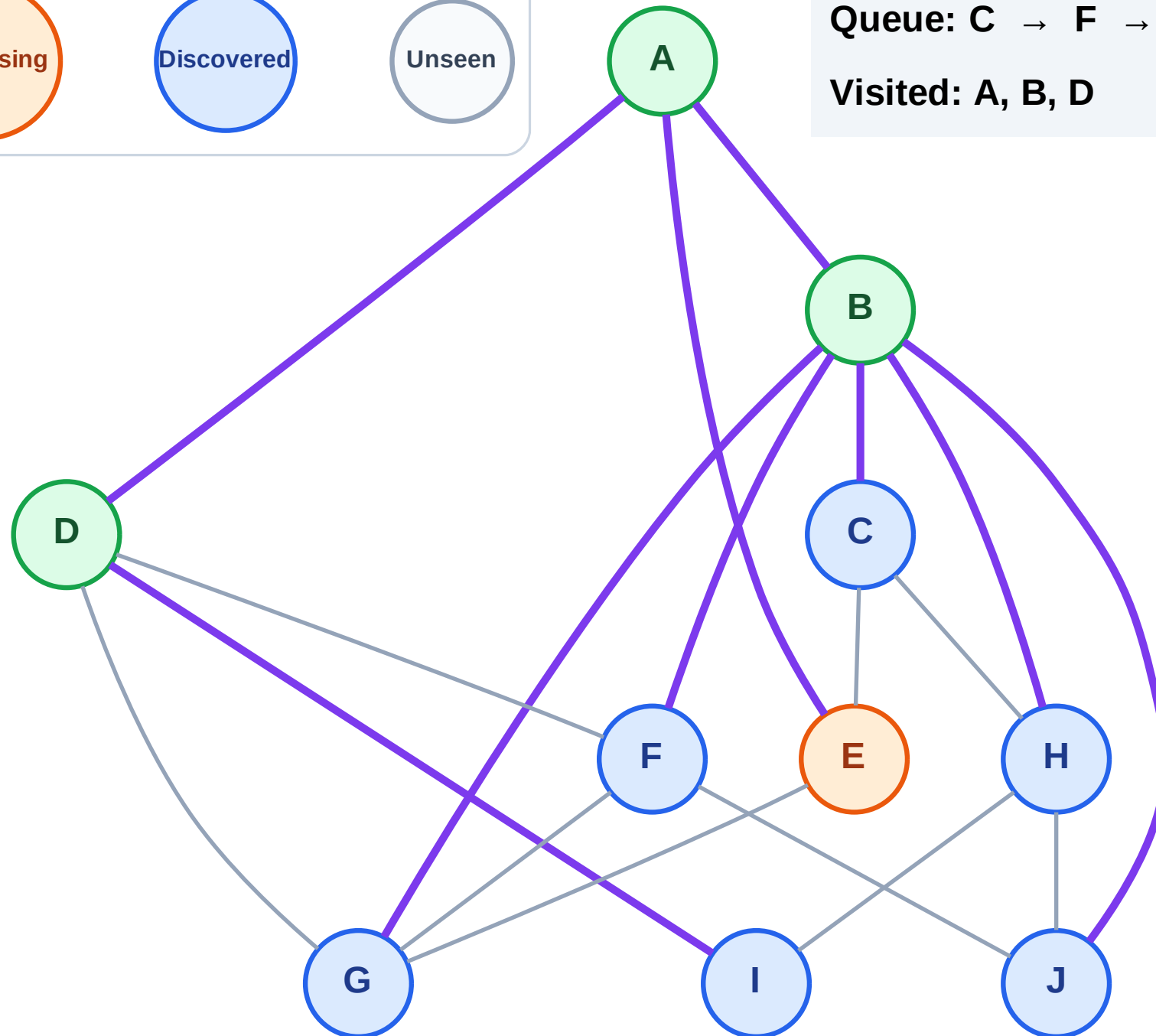
Step 8: Process node E

Legend



Queue: C → F → G → H → J → I

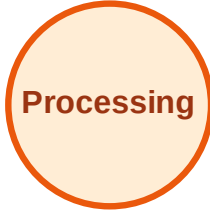
Visited: A, B, D



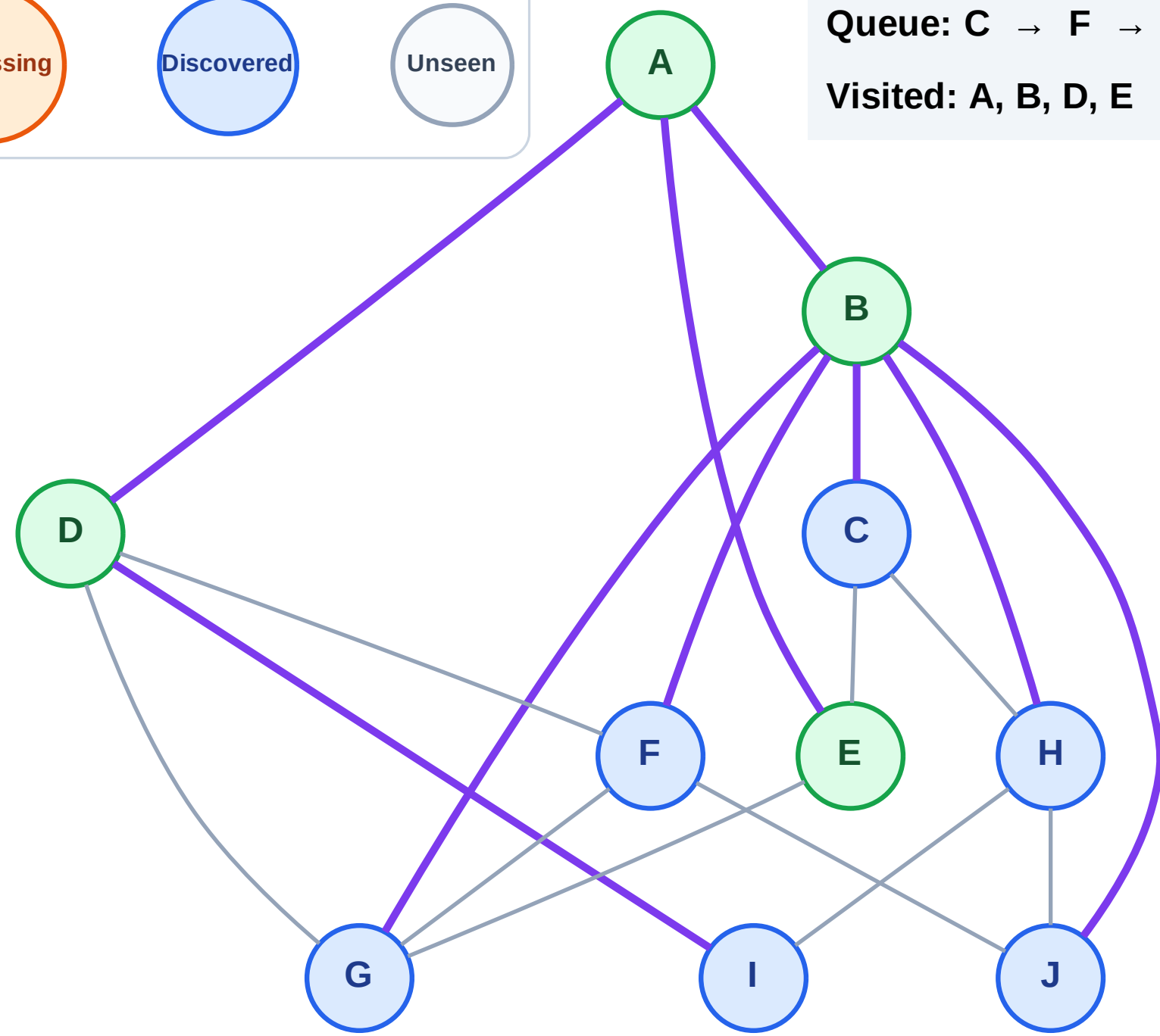
BFS Traversal

Step 9: No new neighbors from E

Legend



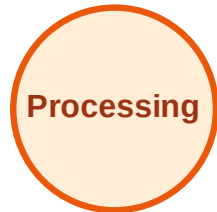
Queue: C → F → G → H → J → I
Visited: A, B, D, E



BFS Traversal

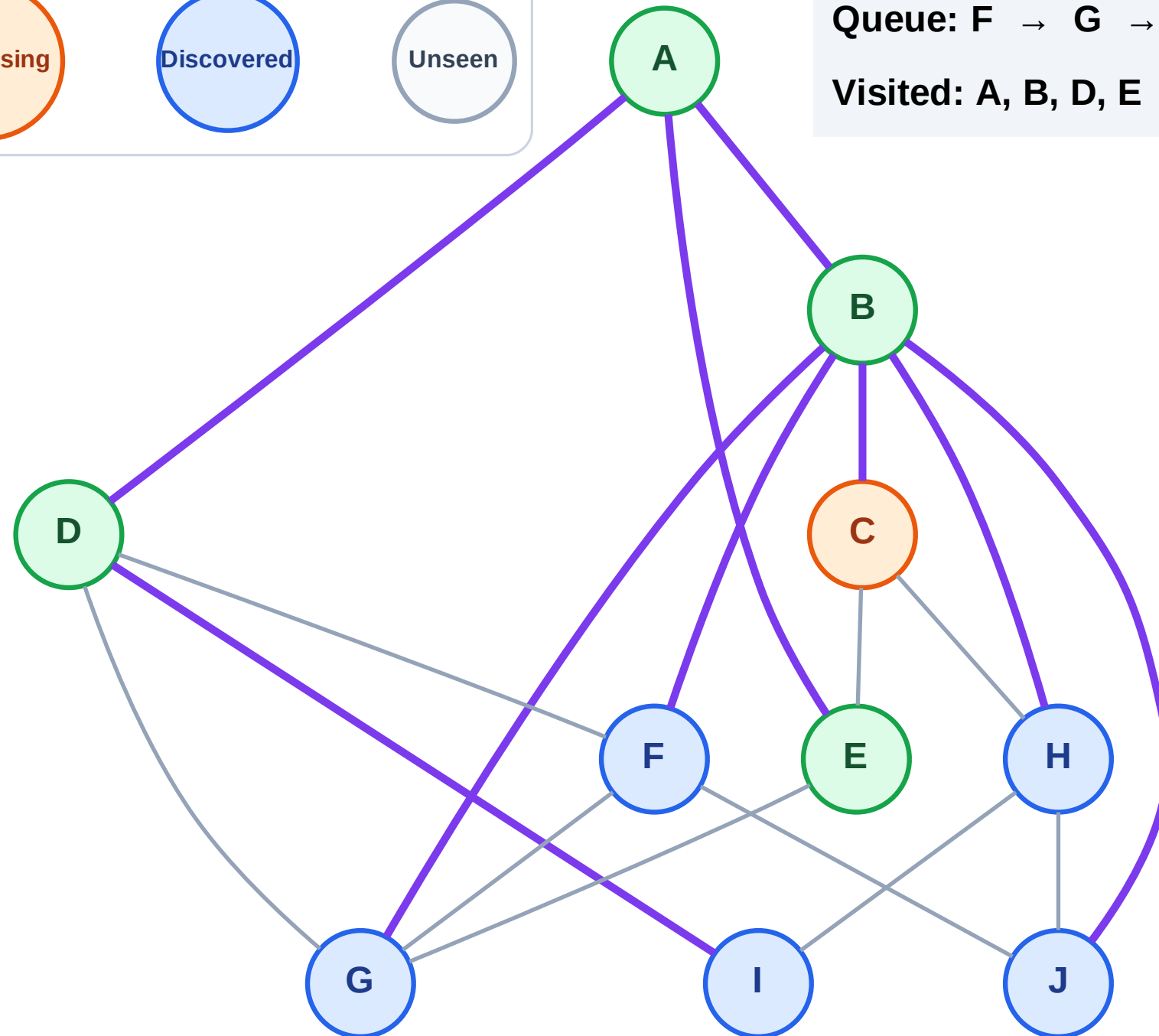
Step 10: Process node C

Legend



Queue: F → G → H → J → I

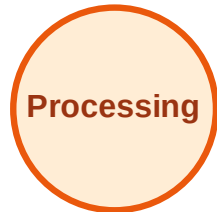
Visited: A, B, D, E



BFS Traversal

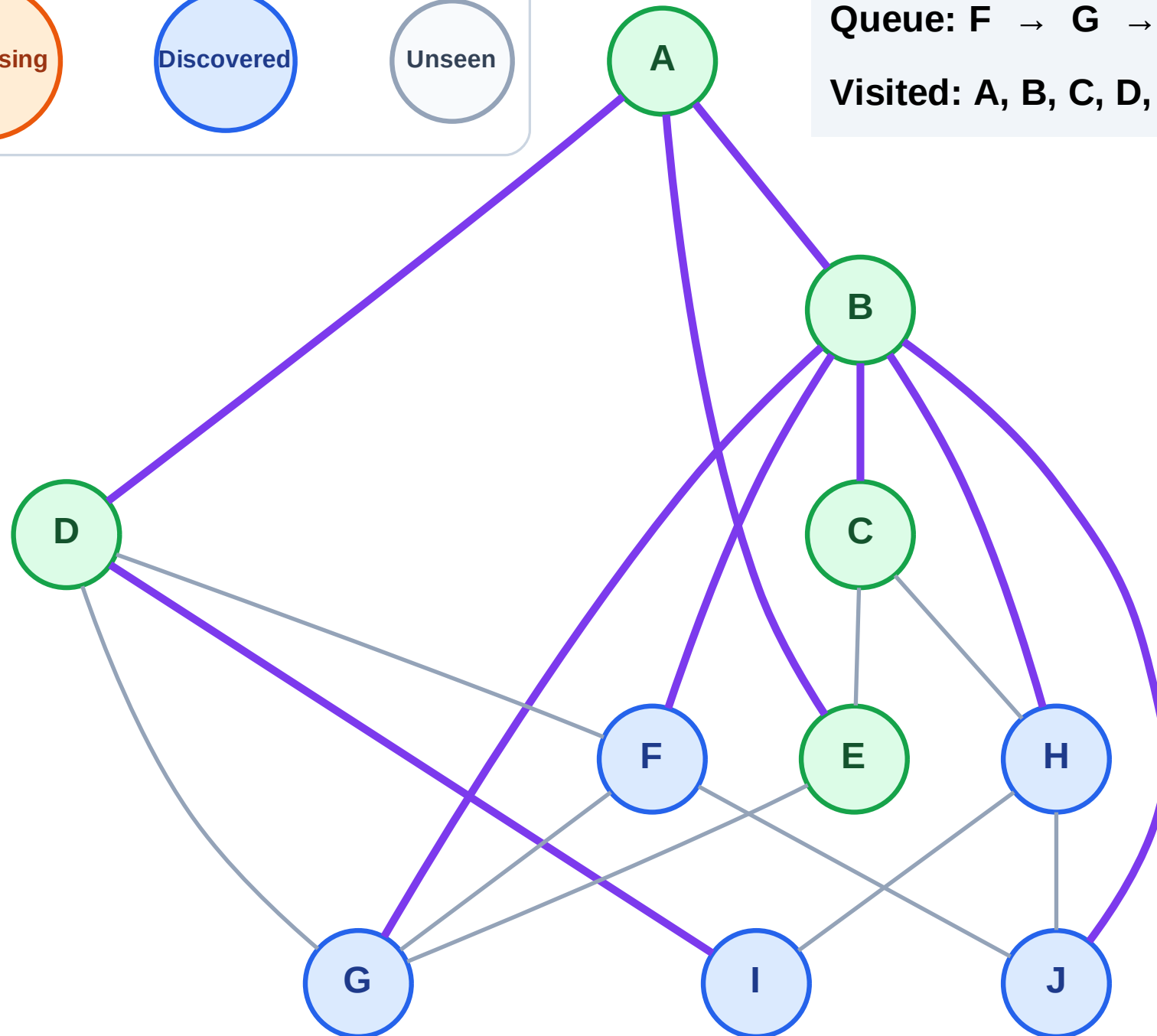
Step 11: No new neighbors from C

Legend



Queue: F → G → H → J → I

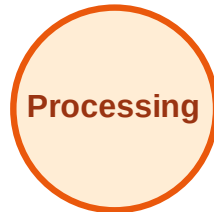
Visited: A, B, C, D, E



BFS Traversal

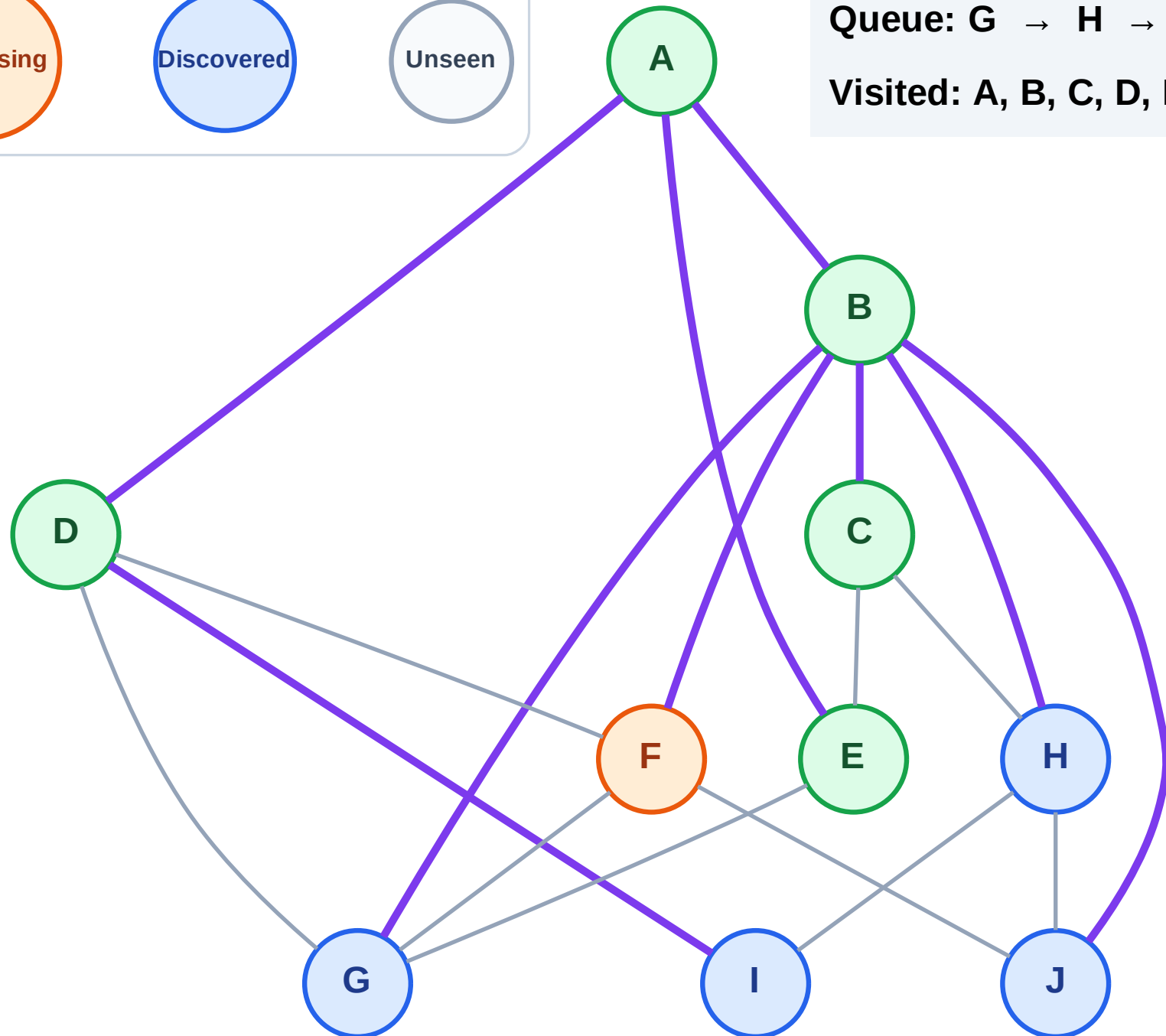
Step 12: Process node F

Legend



Queue: G → H → J → I

Visited: A, B, C, D, E



BFS Traversal

Step 13: No new neighbors from F

Legend

Complete

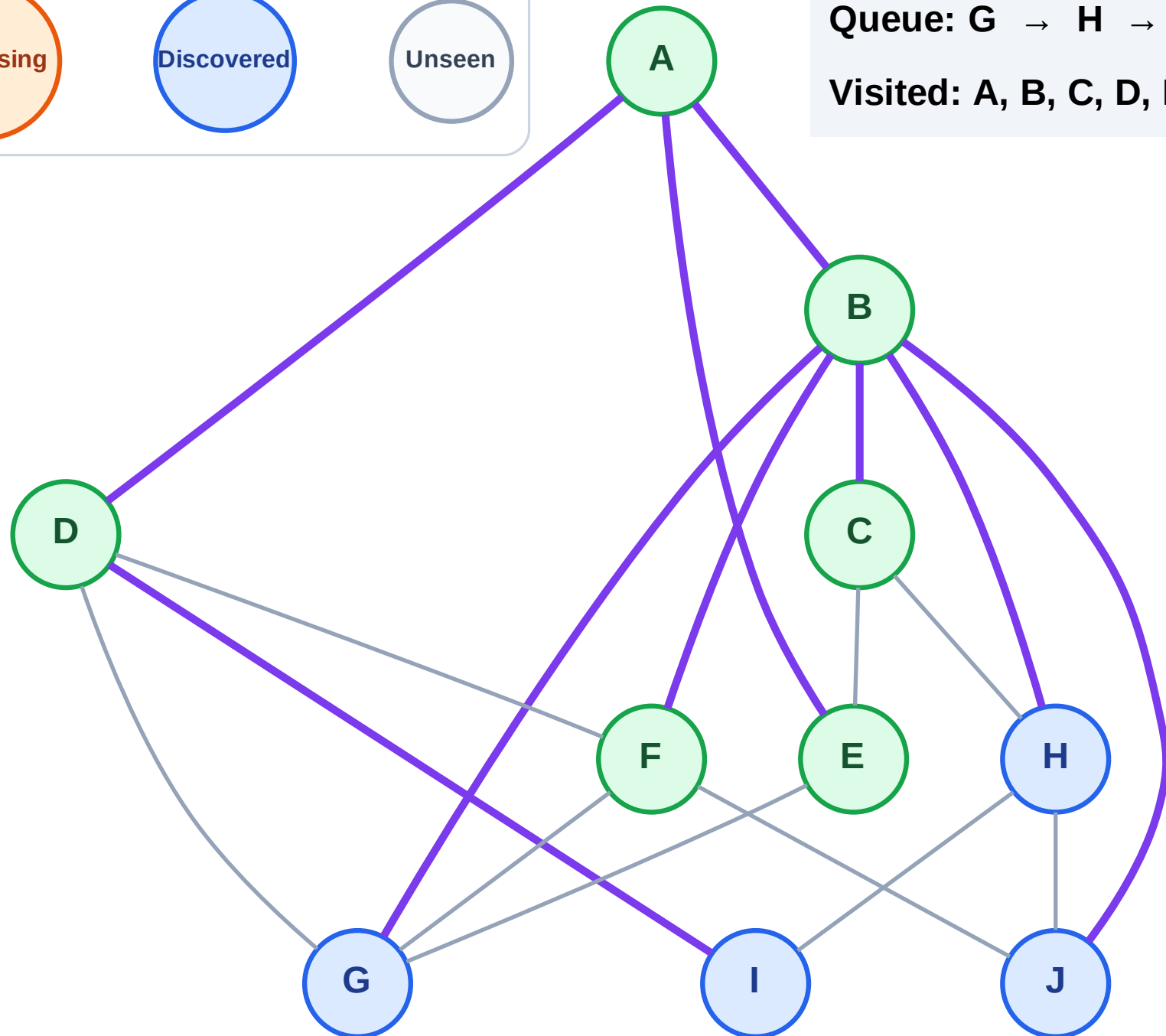
Processing

Discovered

Unseen

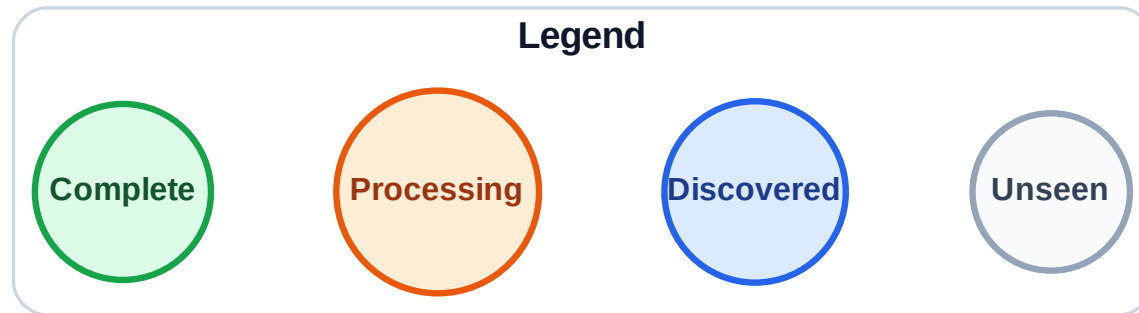
Queue: G → H → J → I

Visited: A, B, C, D, E, F

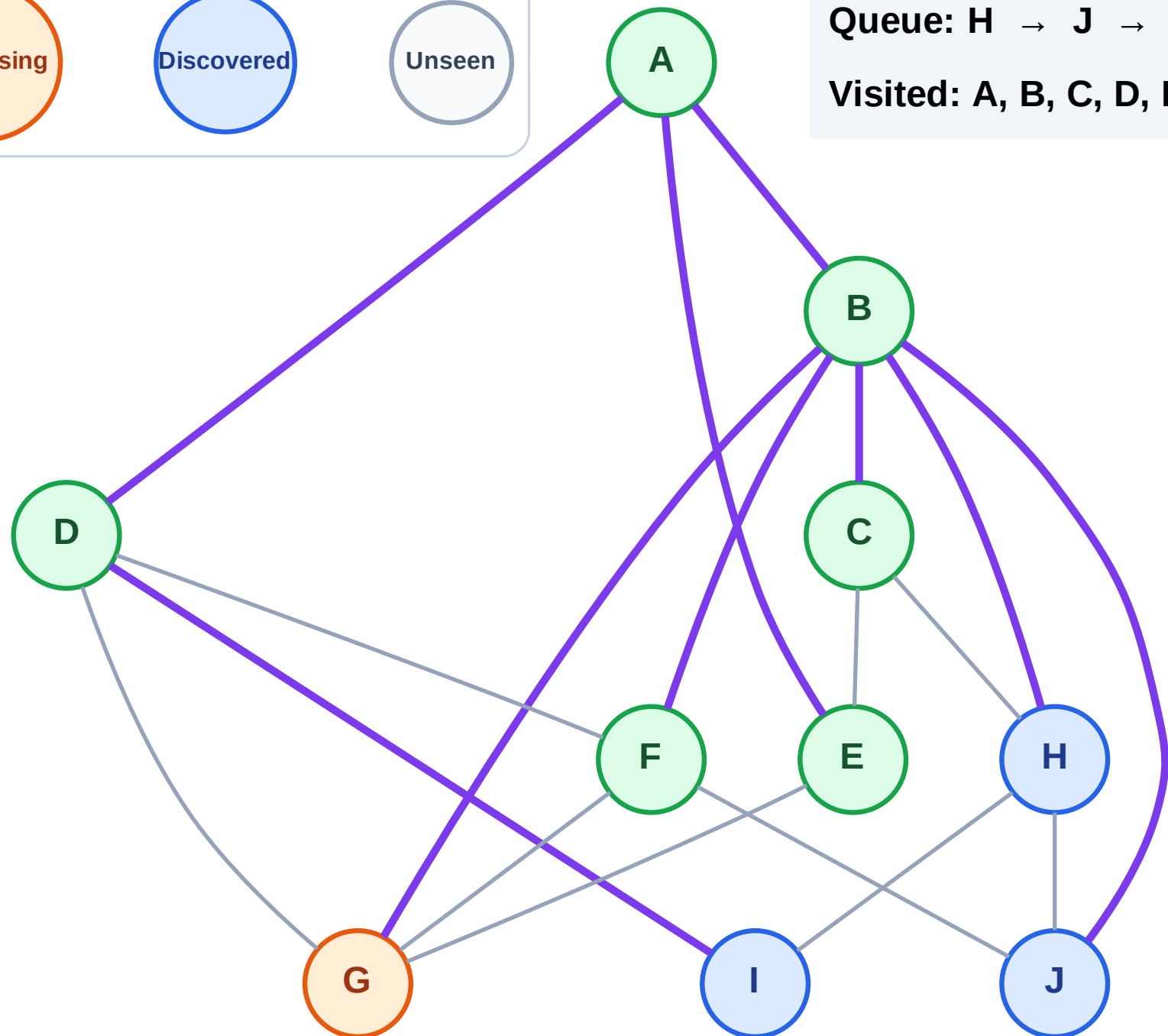


BFS Traversal

Step 14: Process node G



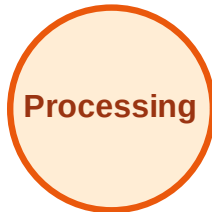
Queue: H → J → I
Visited: A, B, C, D, E, F



BFS Traversal

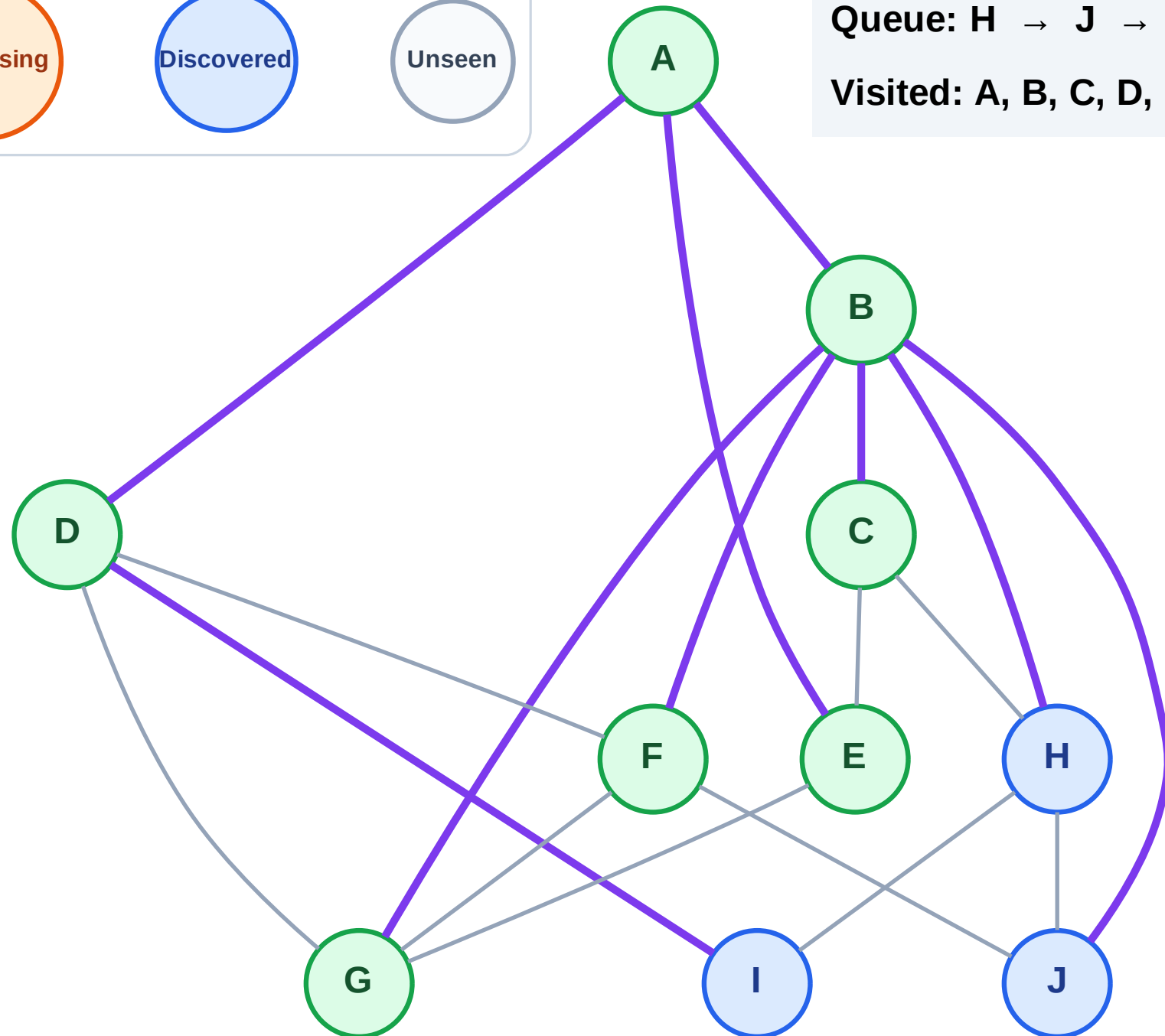
Step 15: No new neighbors from G

Legend



Queue: H → J → I

Visited: A, B, C, D, E, F, G



BFS Traversal

Step 16: Process node H

Legend

Complete

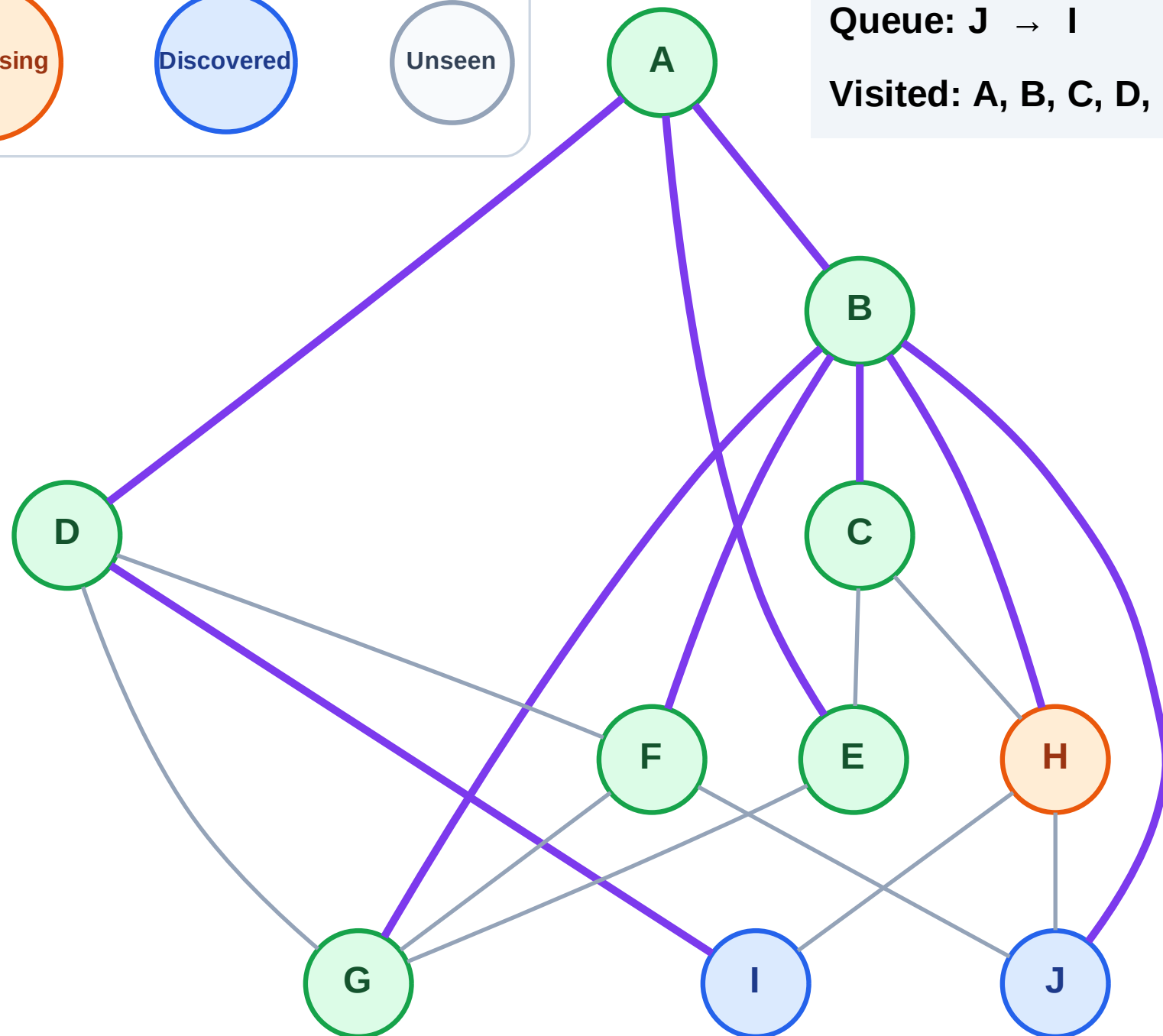
Processing

Discovered

Unseen

Queue: J → I

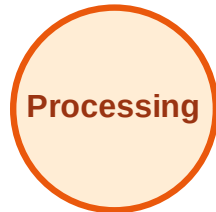
Visited: A, B, C, D, E, F, G



BFS Traversal

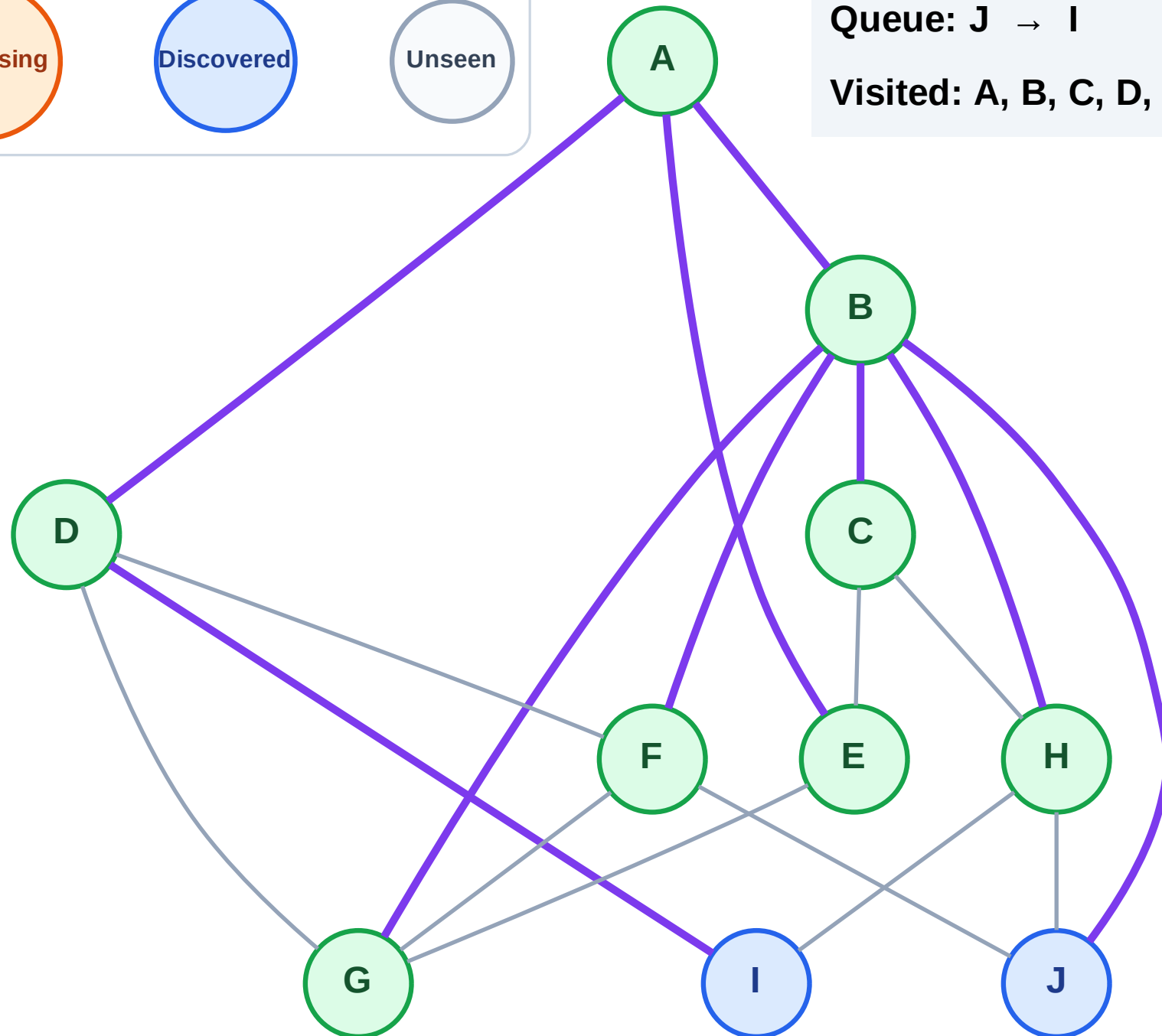
Step 17: No new neighbors from H

Legend



Queue: J → I

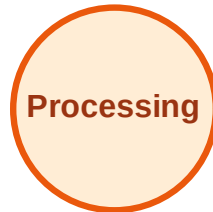
Visited: A, B, C, D, E, F, G, H



BFS Traversal

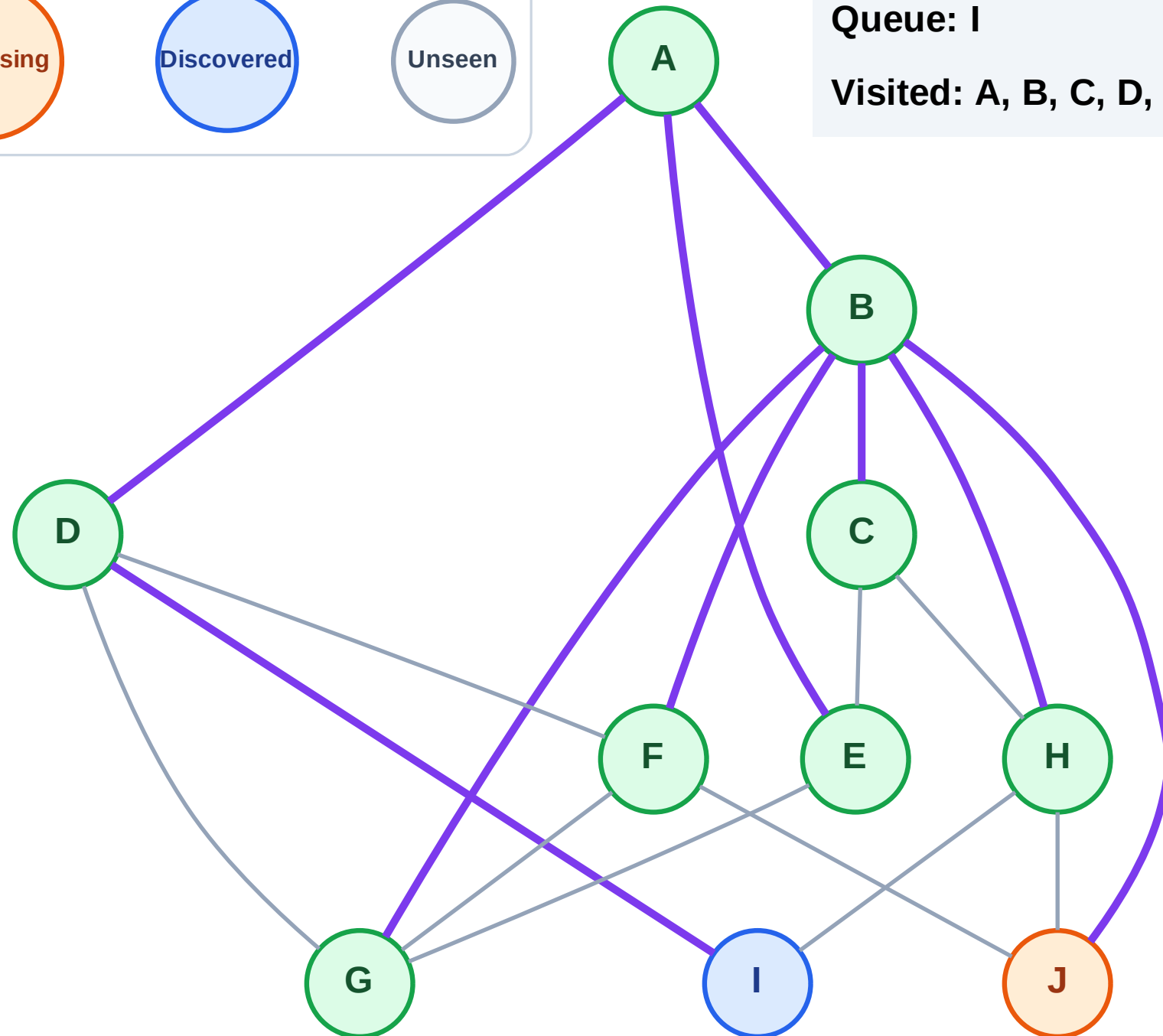
Step 18: Process node J

Legend



Queue: I

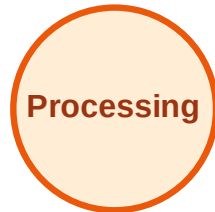
Visited: A, B, C, D, E, F, G, H



BFS Traversal

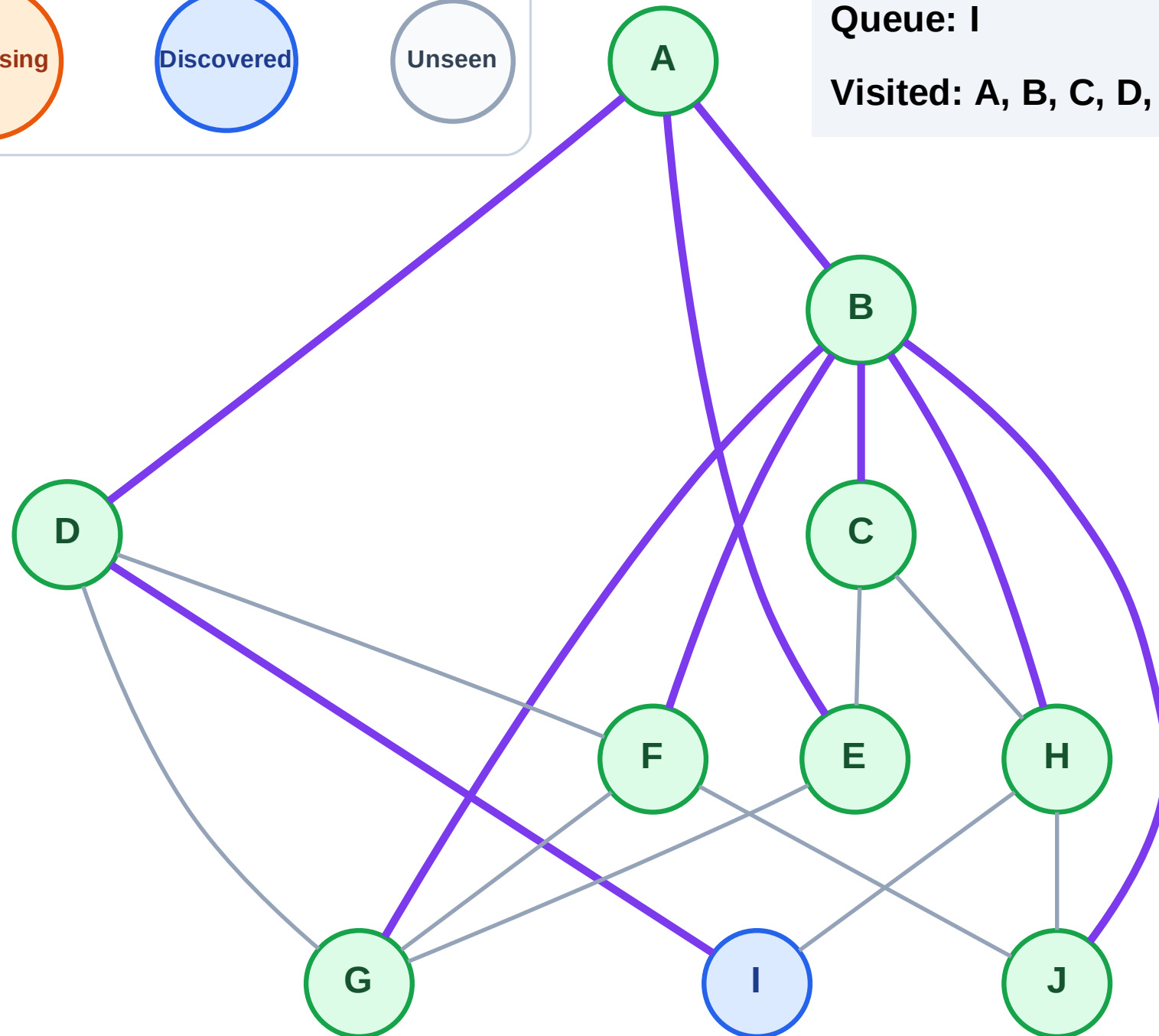
Step 19: No new neighbors from J

Legend



Queue: I

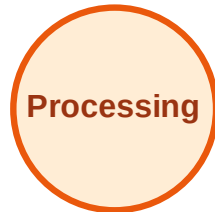
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

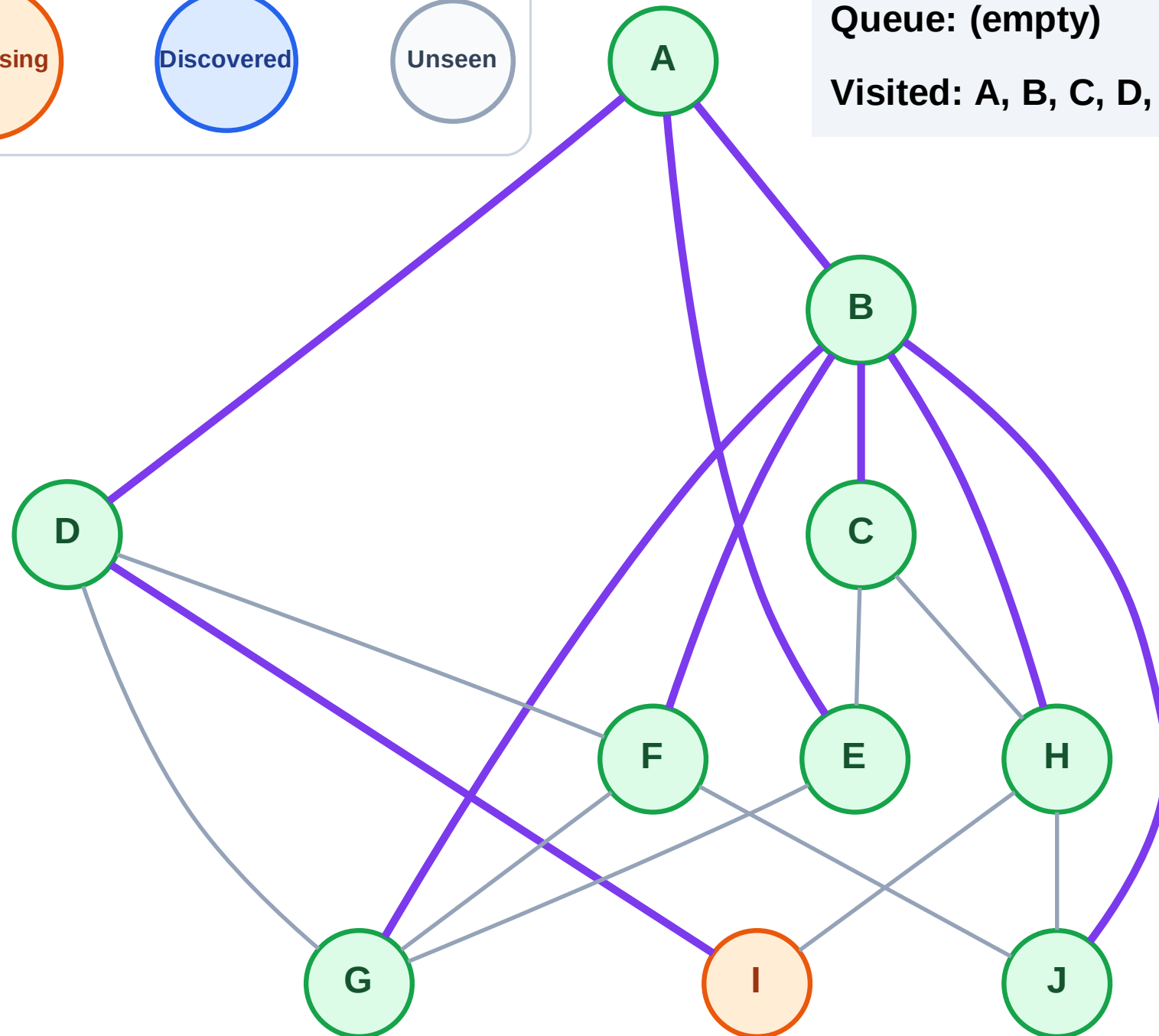
Step 20: Process node I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

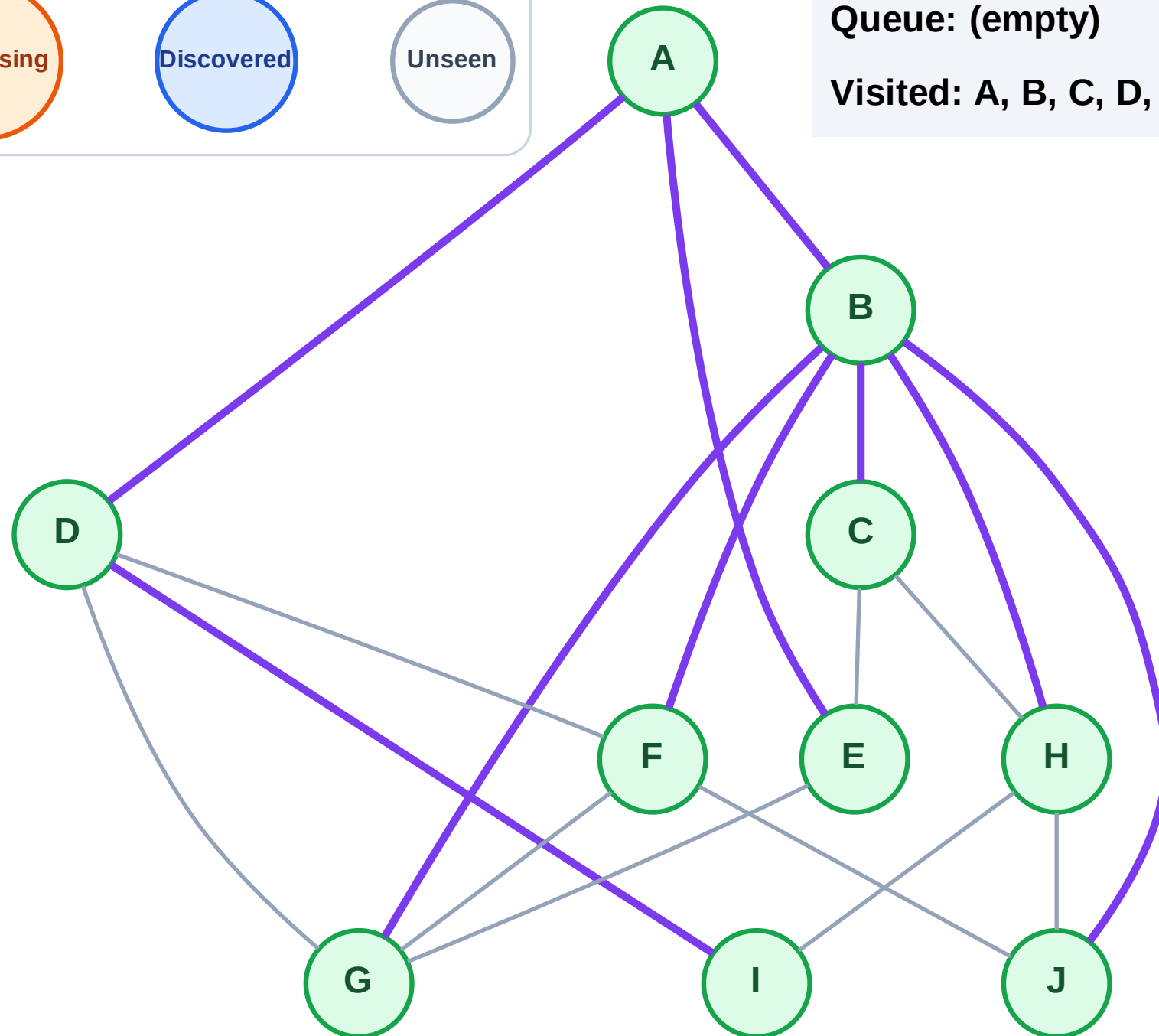
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

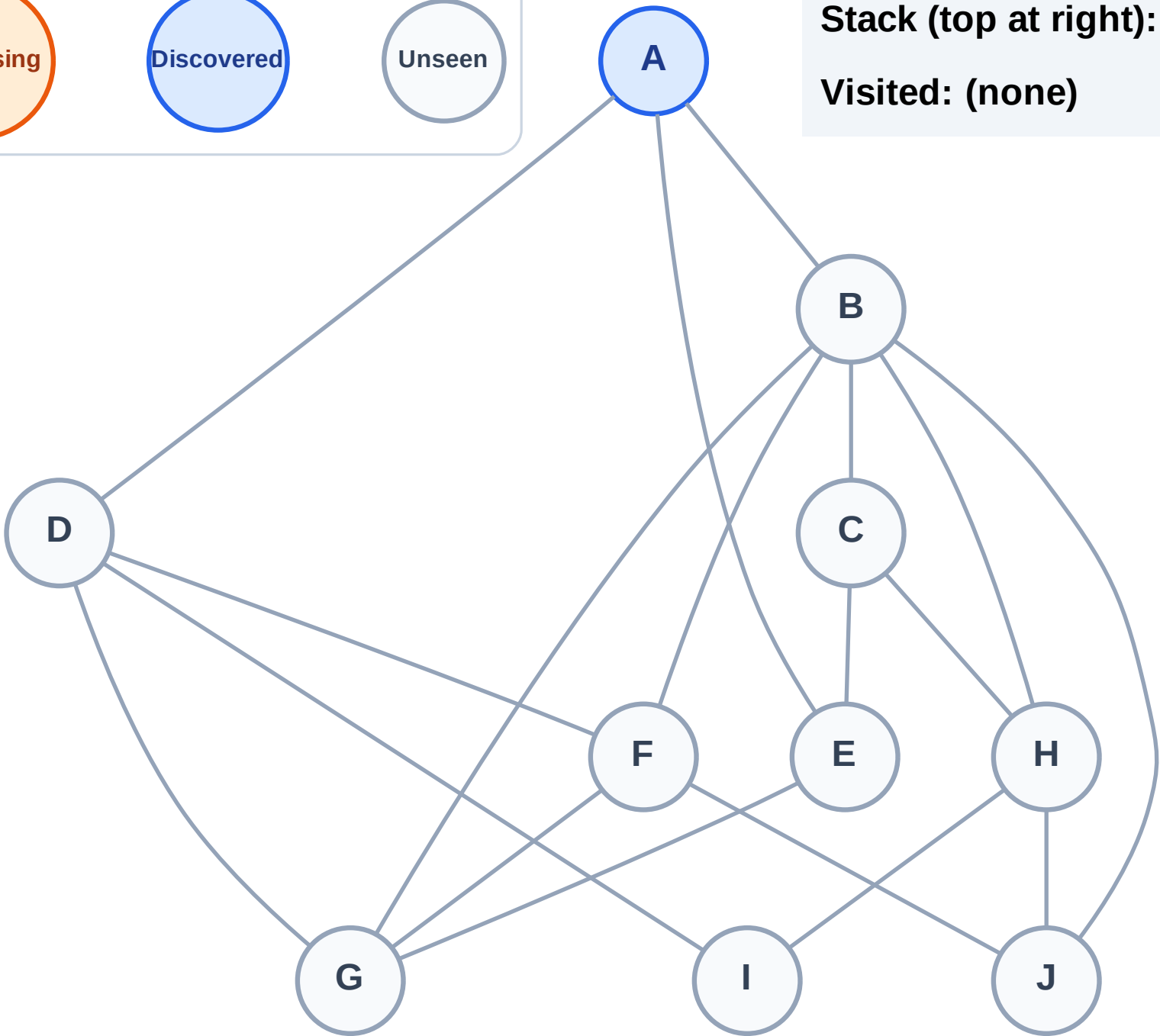
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

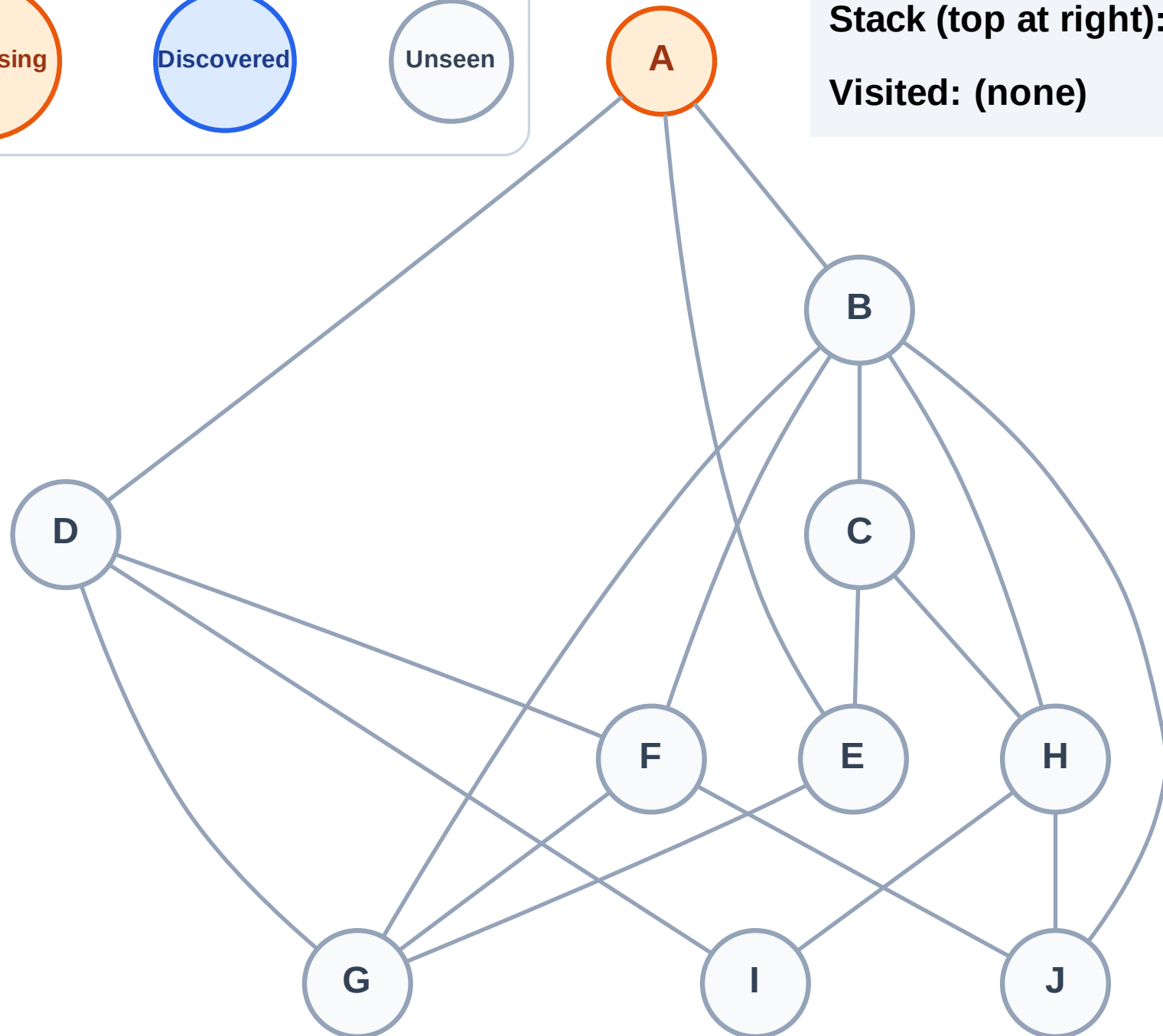
Processing

Discovered

Unseen

Stack (top at right): (empty)

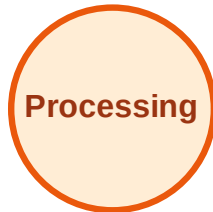
Visited: (none)



DFS Traversal

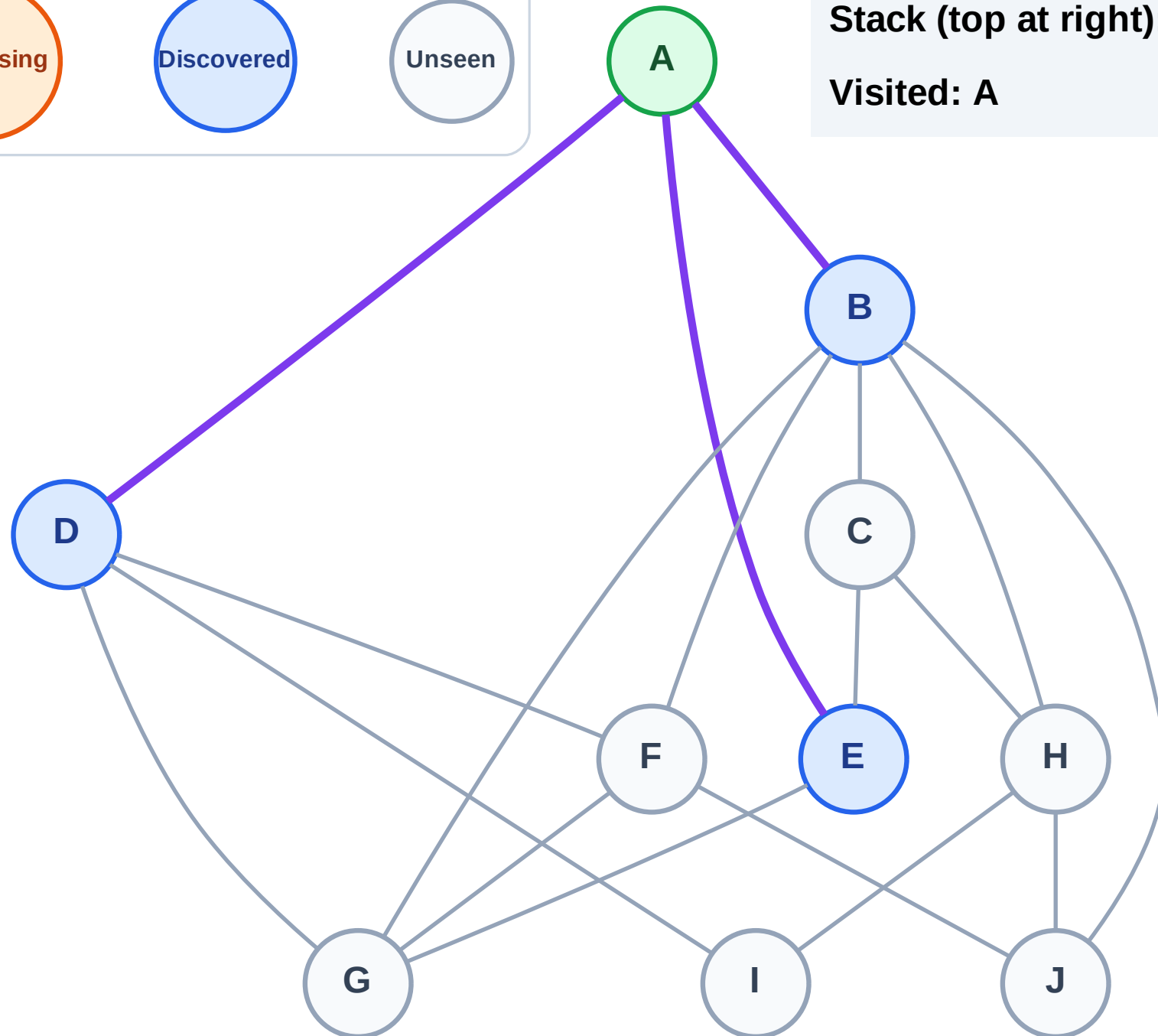
Step 3: Discover E, D, B from A

Legend



Stack (top at right): E | D | B

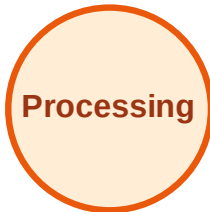
Visited: A



DFS Traversal

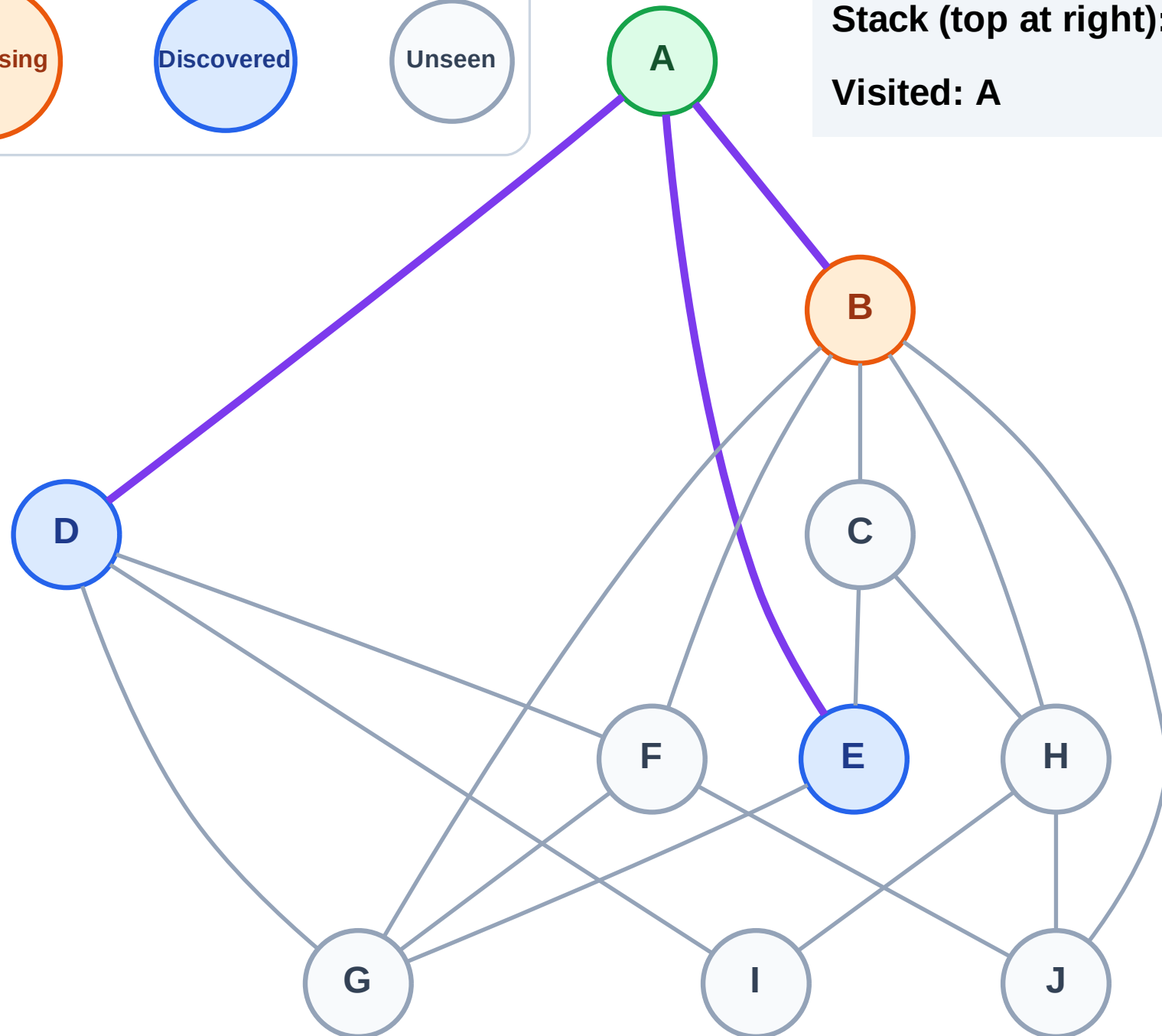
Step 4: Process node B

Legend



Stack (top at right): E | D

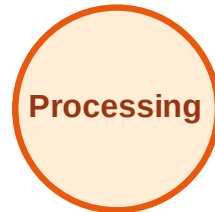
Visited: A



DFS Traversal

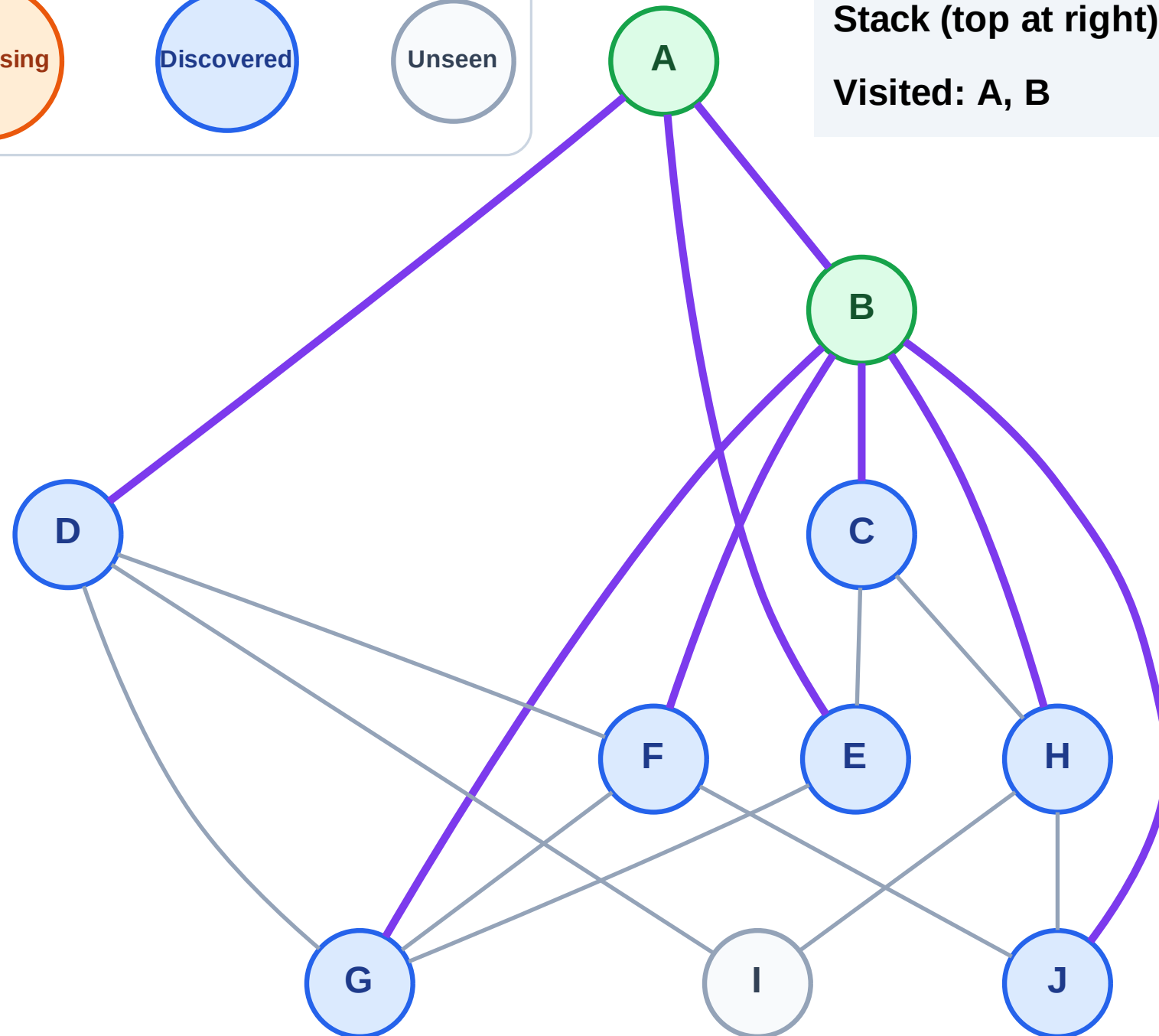
Step 5: Discover J, H, G, F, C from B

Legend

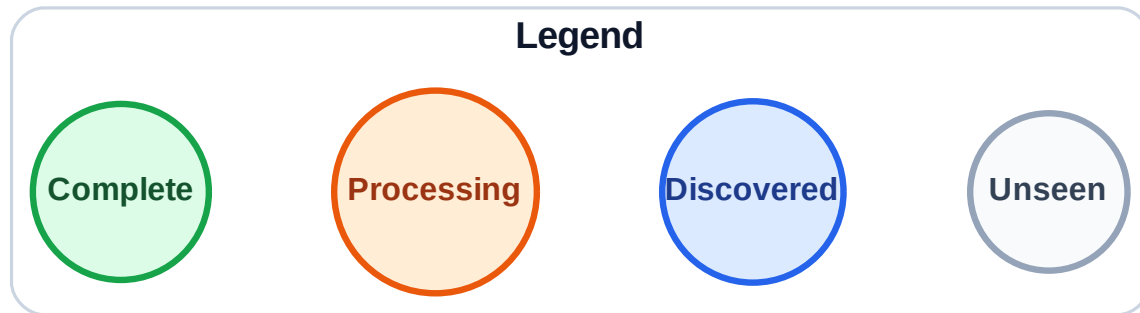


Stack (top at right): E | D | J | H | G | F | C

Visited: A, B

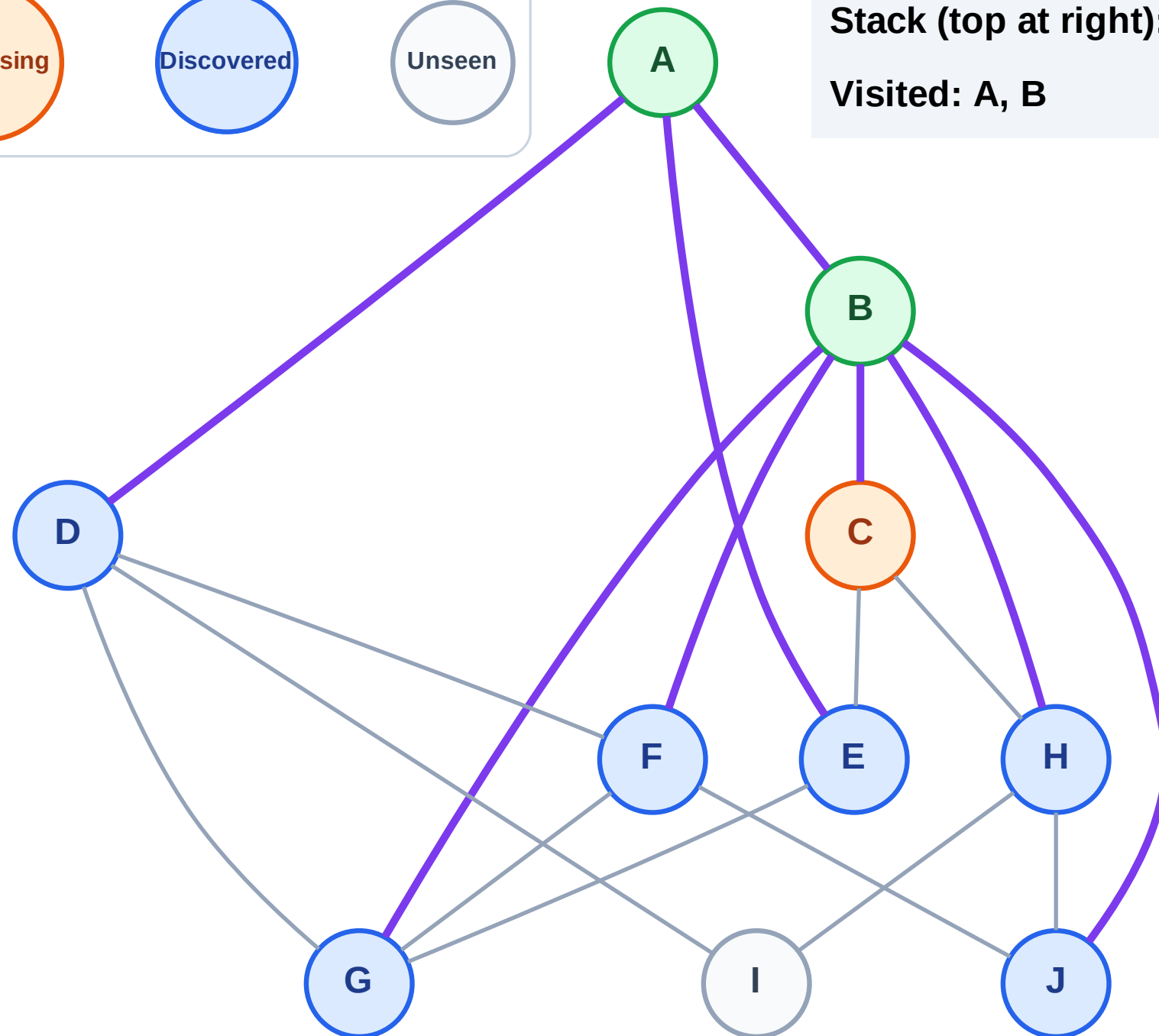


Step 6: Process node C



Stack (top at right): E | D | J | H | G | F

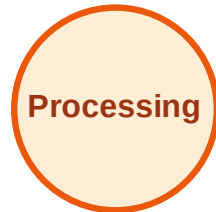
Visited: A, B



DFS Traversal

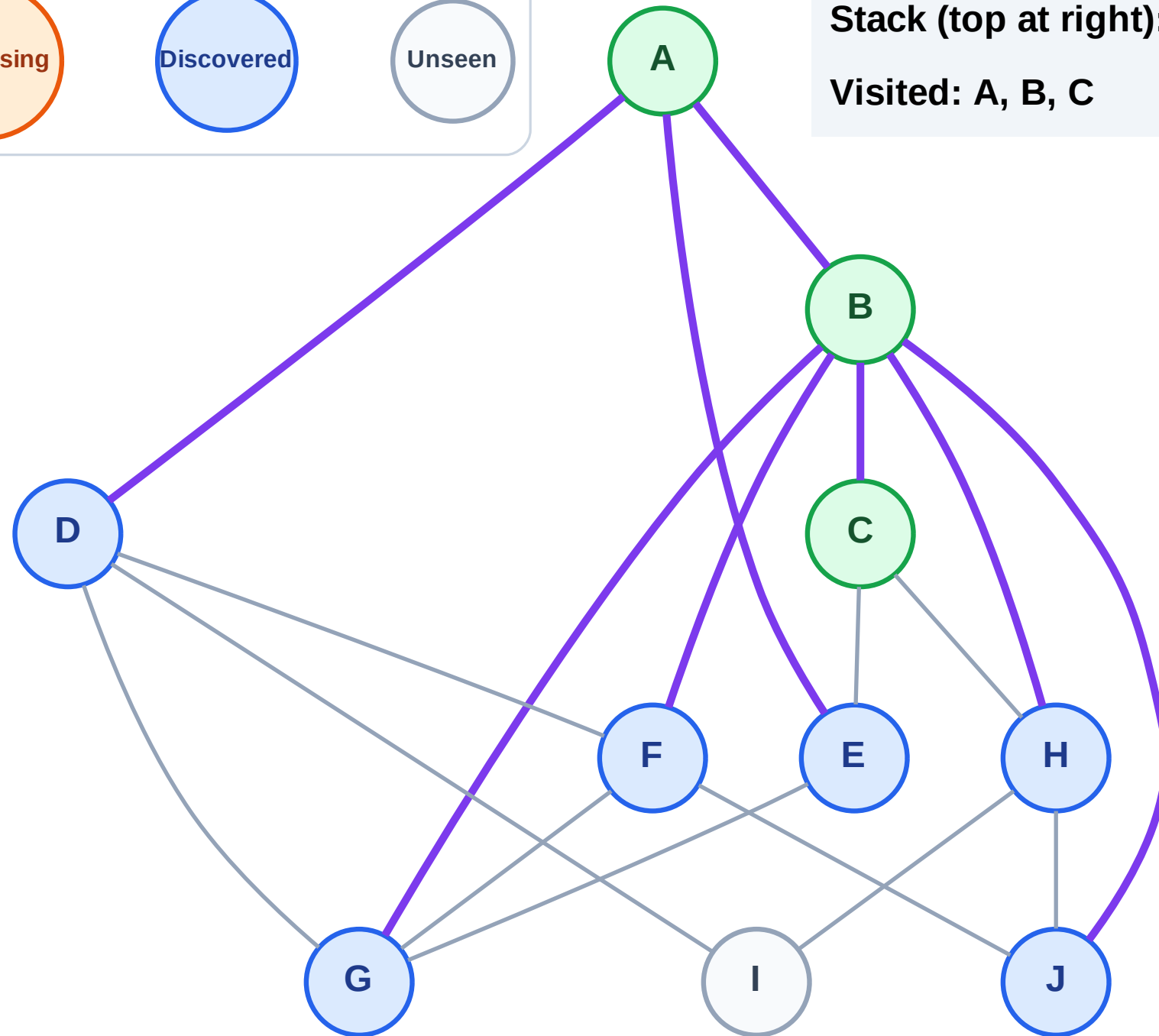
Step 7: No new neighbors from C

Legend



Stack (top at right): E | D | J | H | G | F

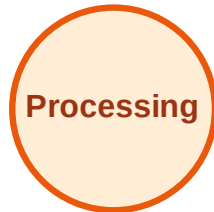
Visited: A, B, C



DFS Traversal

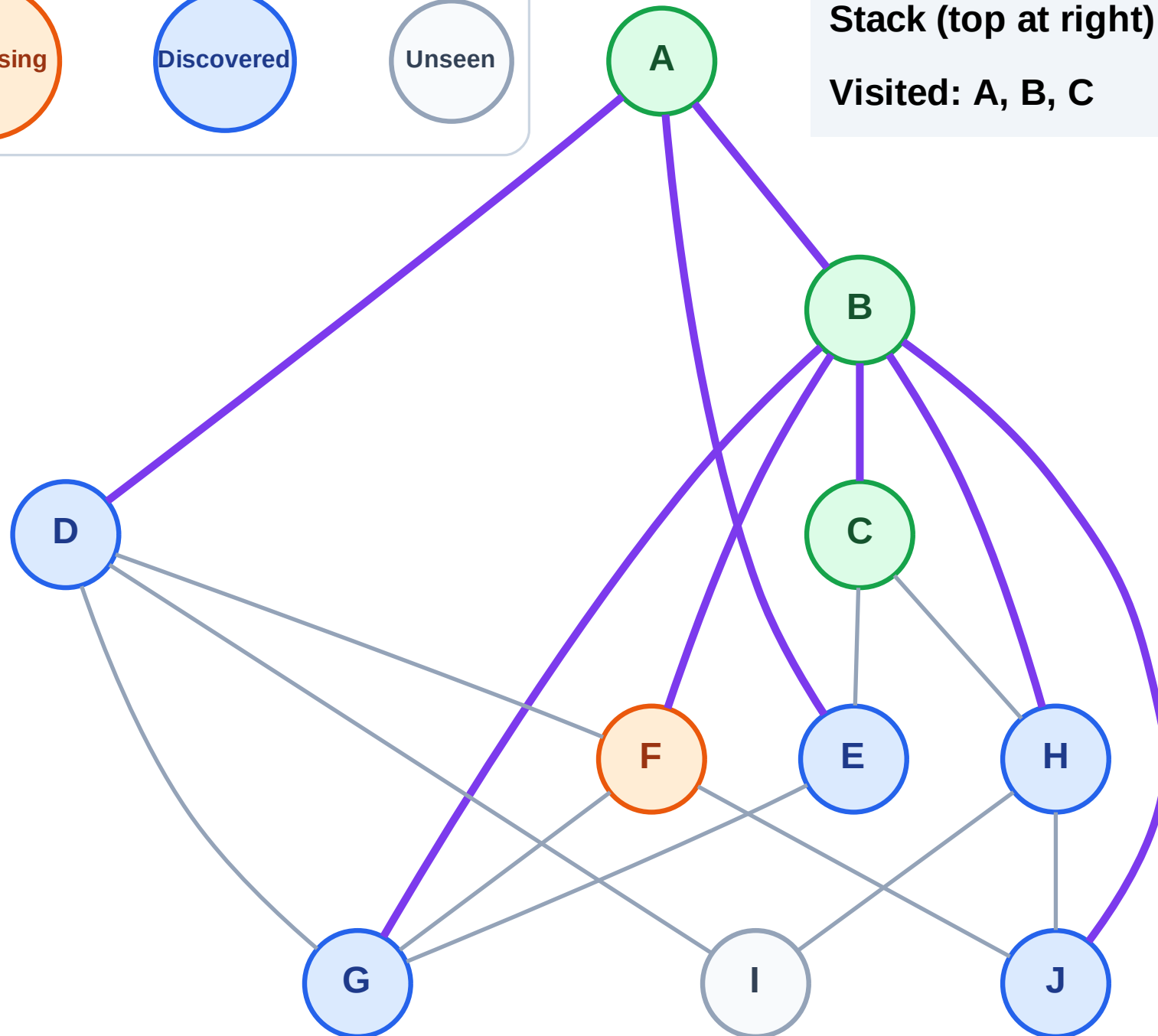
Step 8: Process node F

Legend



Stack (top at right): E | D | J | H | G

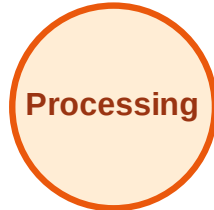
Visited: A, B, C



DFS Traversal

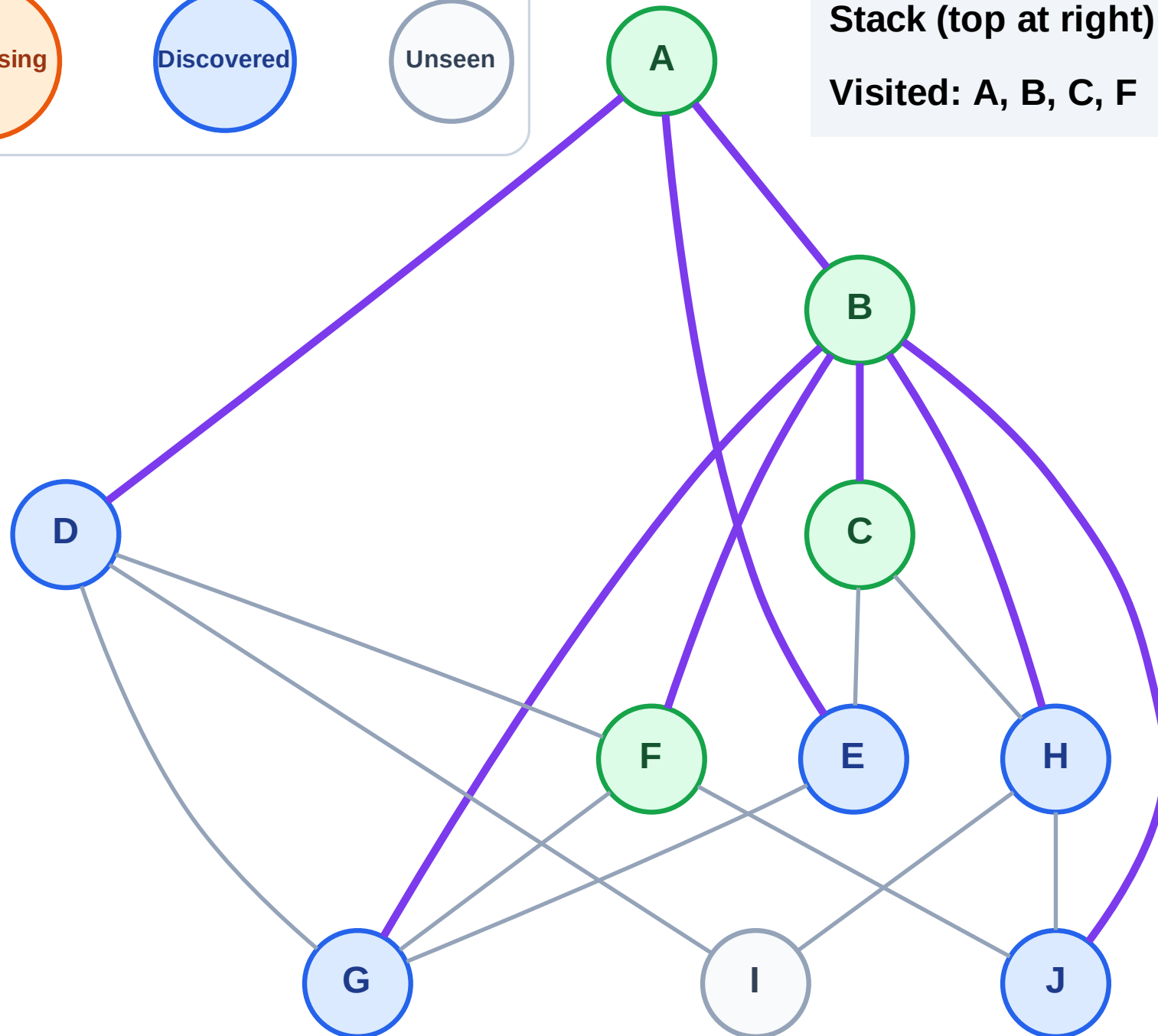
Step 9: No new neighbors from F

Legend



Stack (top at right): E | D | J | H | G

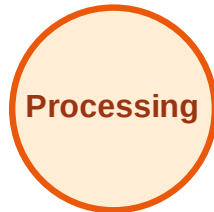
Visited: A, B, C, F



DFS Traversal

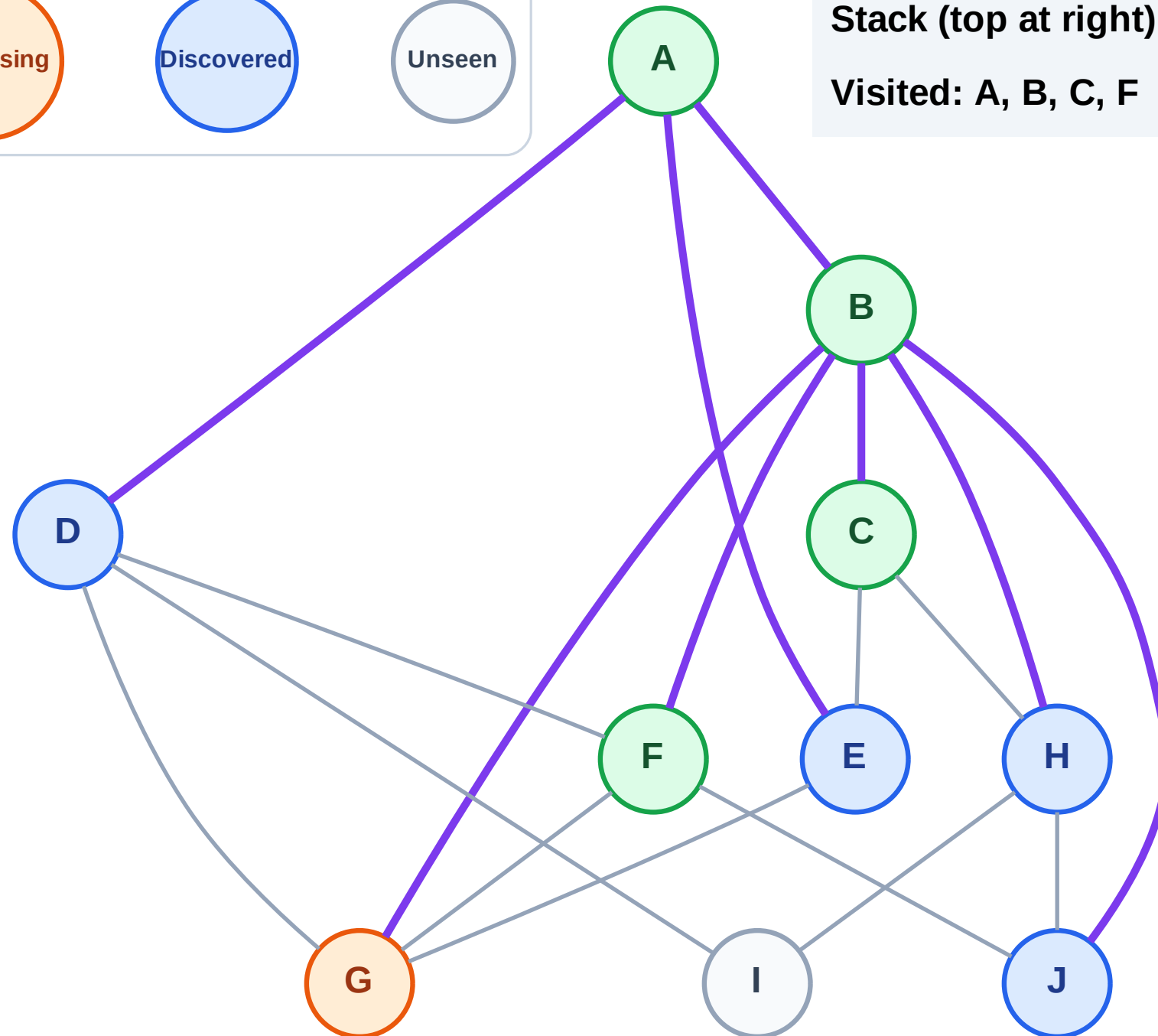
Step 10: Process node G

Legend



Stack (top at right): E | D | J | H

Visited: A, B, C, F



DFS Traversal

Step 11: No new neighbors from G

Legend

Complete

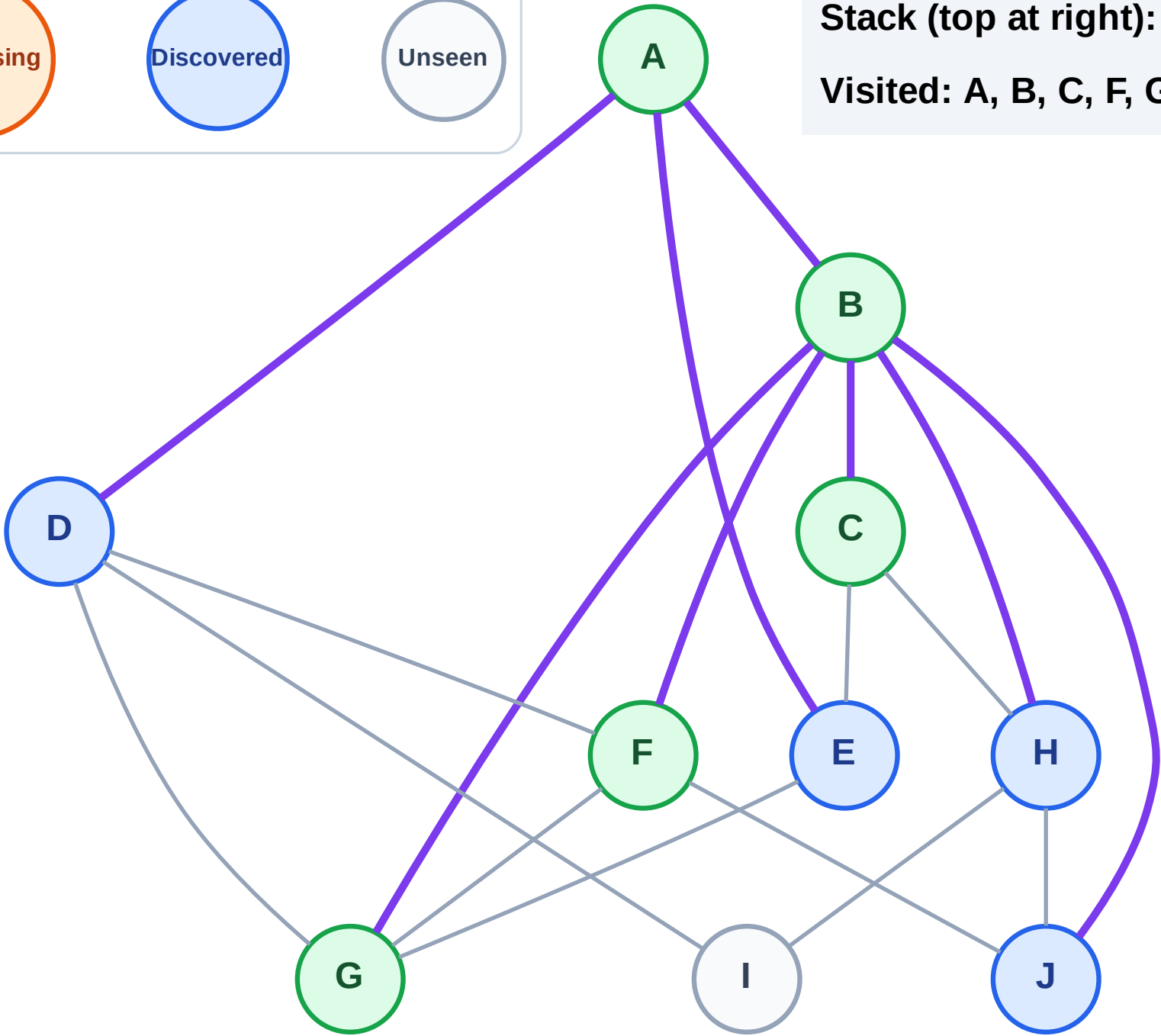
Processing

Discovered

Unseen

Stack (top at right): E | D | J | H

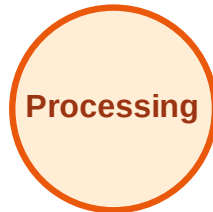
Visited: A, B, C, F, G



DFS Traversal

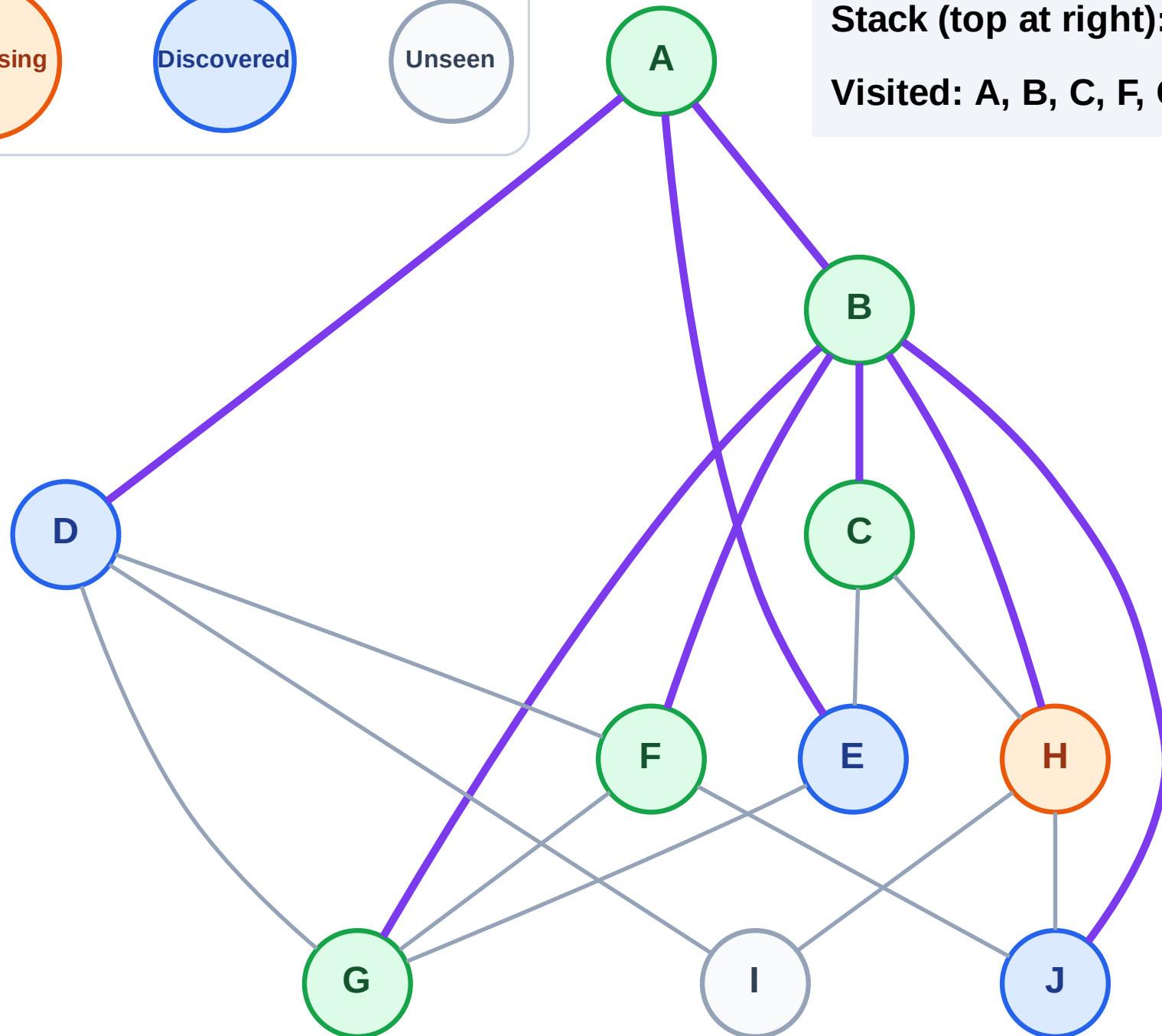
Step 12: Process node H

Legend



Stack (top at right): E | D | J

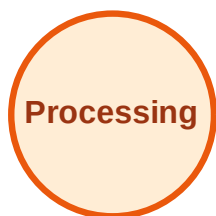
Visited: A, B, C, F, G



DFS Traversal

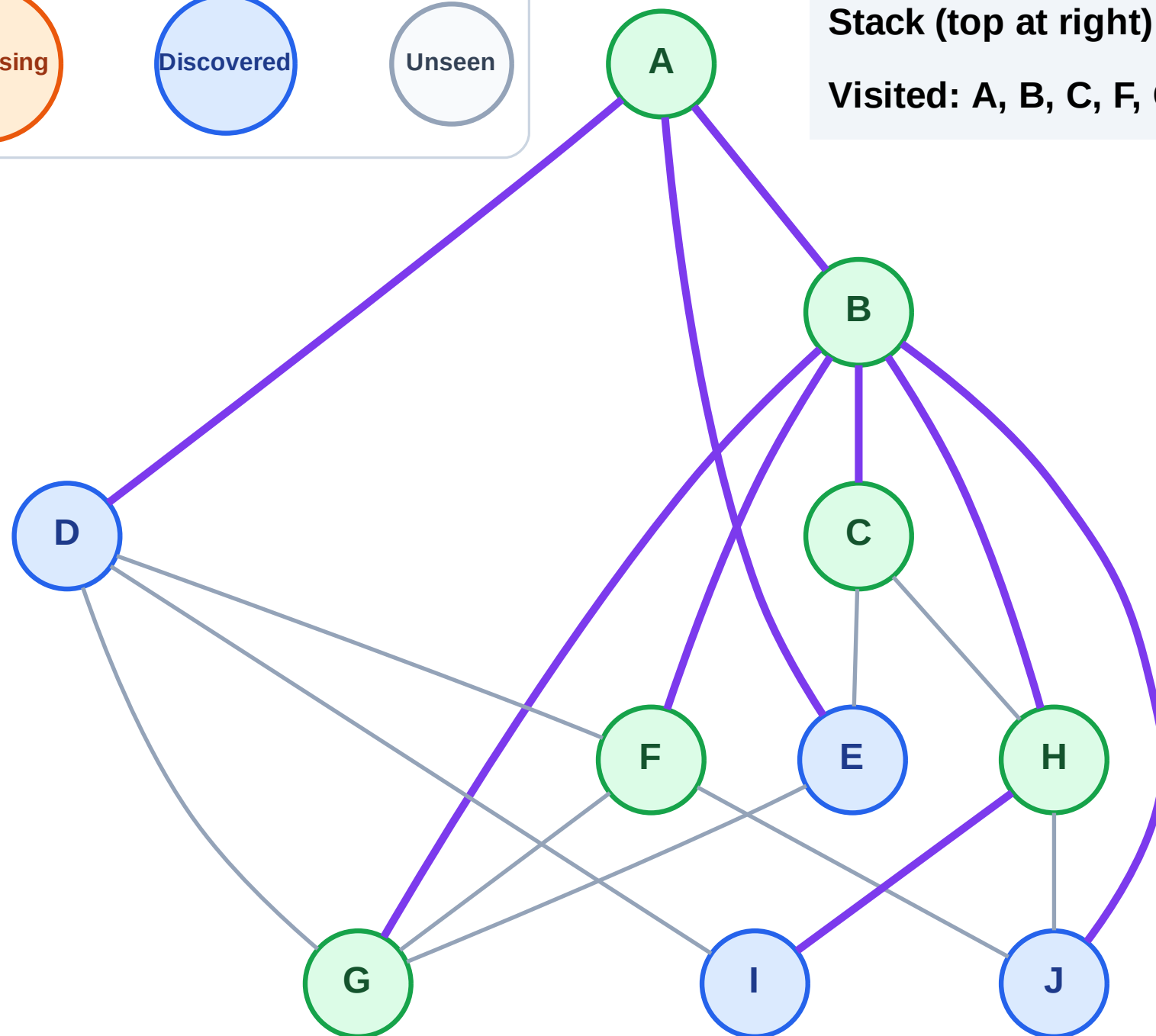
Step 13: Discover I from H

Legend



Stack (top at right): E | D | J | I

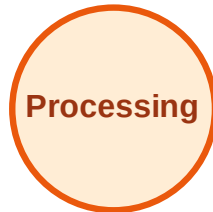
Visited: A, B, C, F, G, H



DFS Traversal

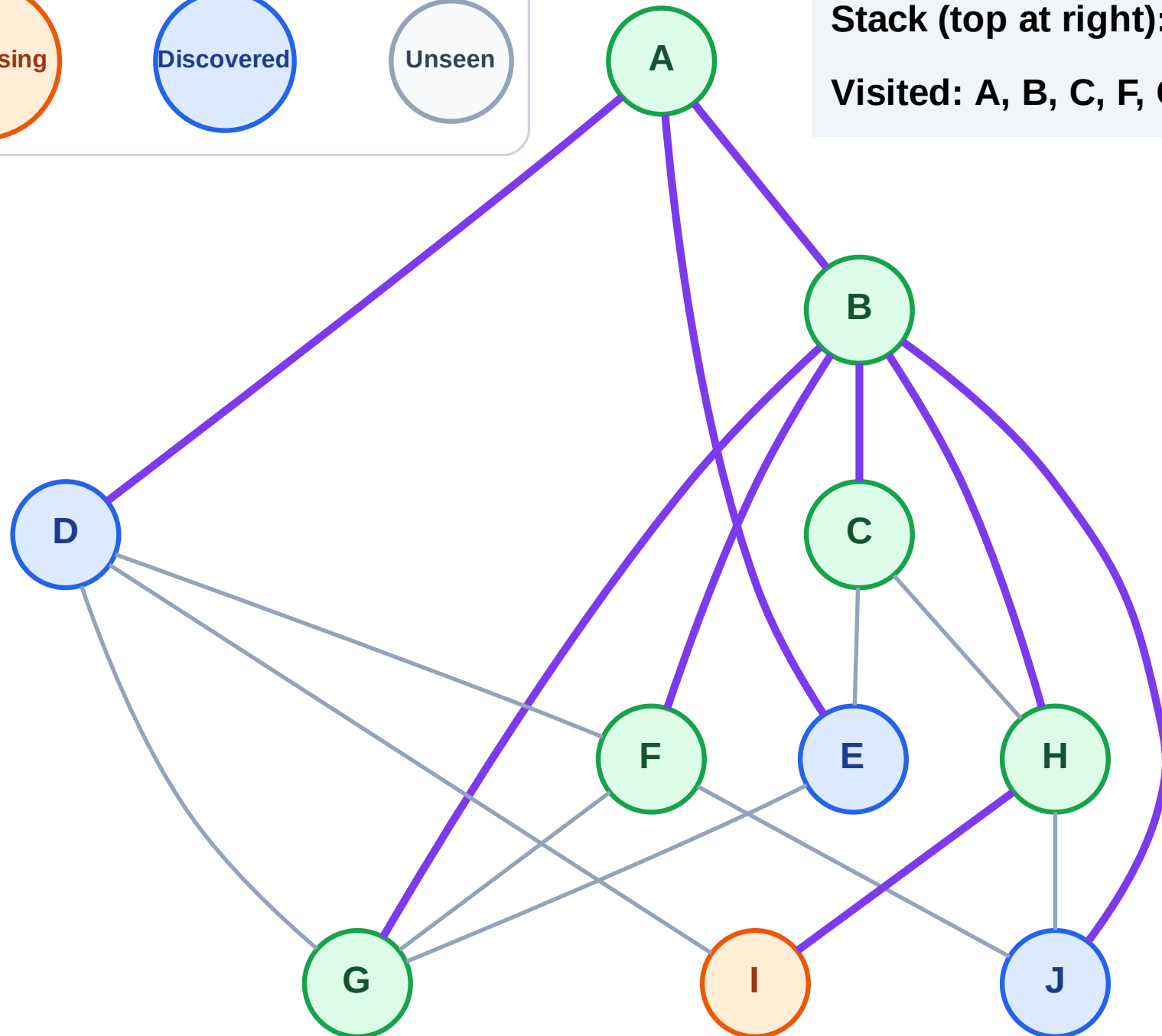
Step 14: Process node I

Legend



Stack (top at right): E | D | J

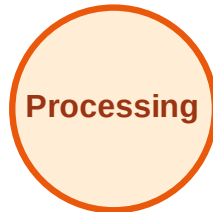
Visited: A, B, C, F, G, H



DFS Traversal

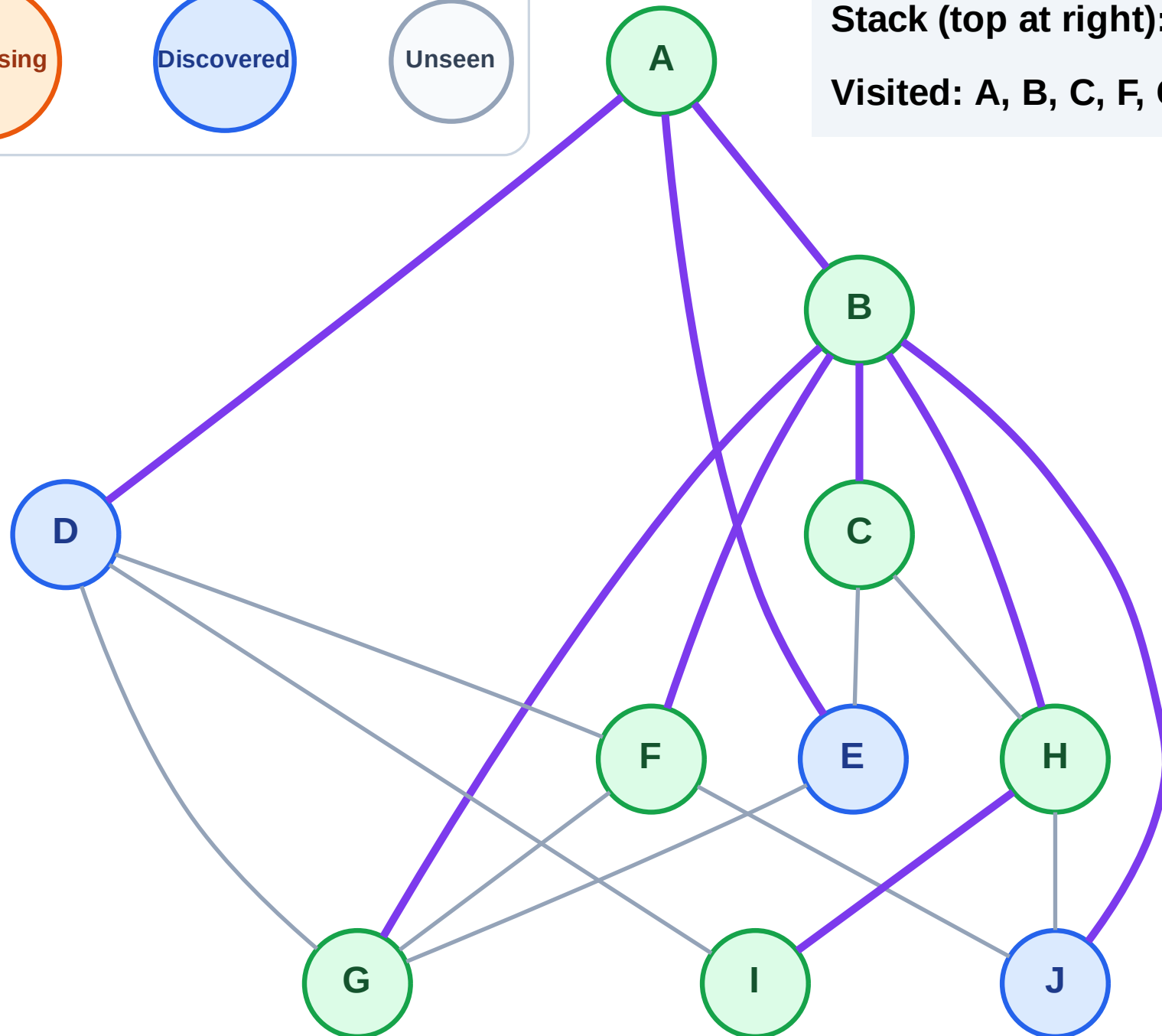
Step 15: No new neighbors from I

Legend



Stack (top at right): E | D | J

Visited: A, B, C, F, G, H, I



DFS Traversal

Step 16: Process node J

Legend

Complete

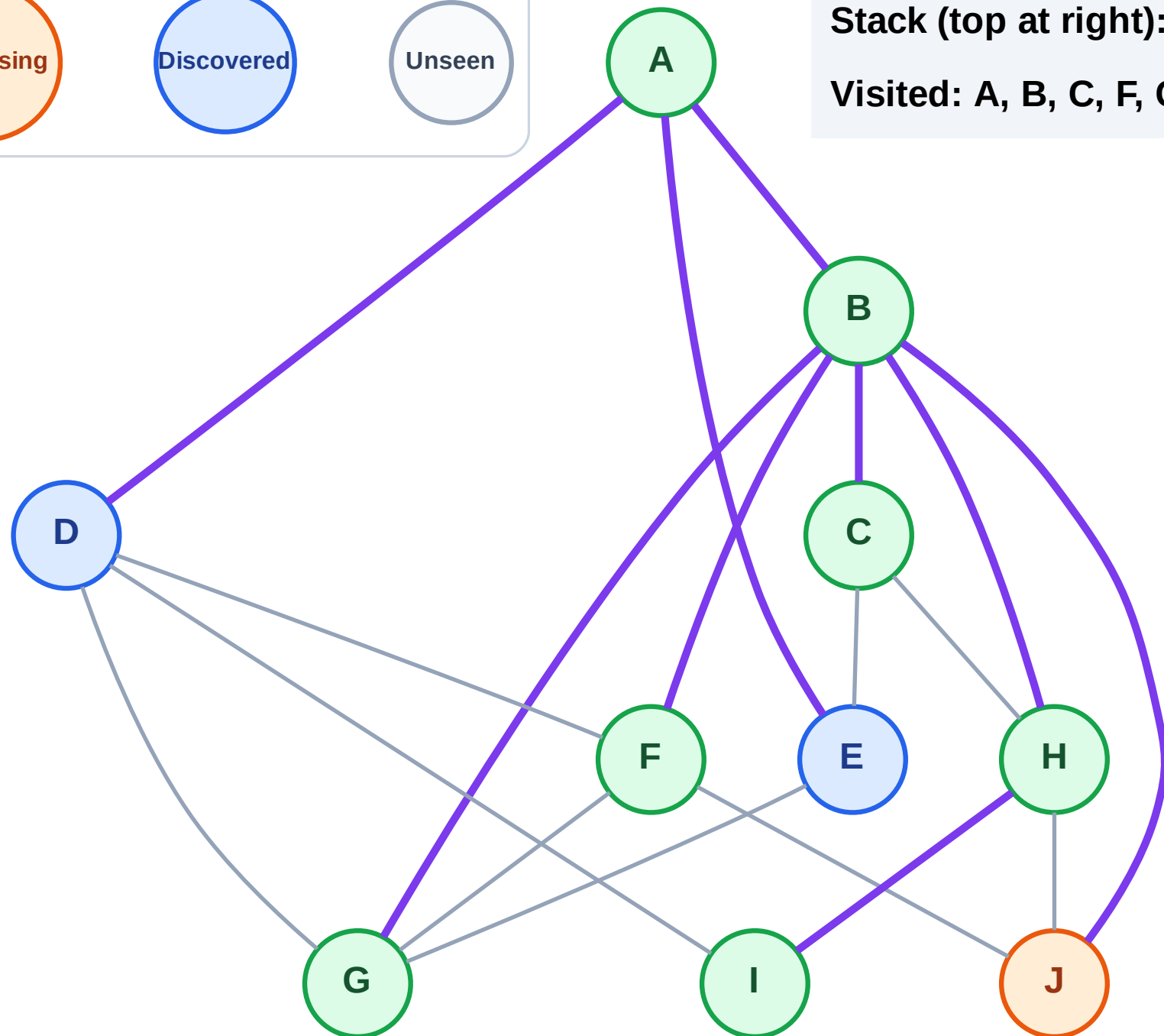
Processing

Discovered

Unseen

Stack (top at right): E | D

Visited: A, B, C, F, G, H, I



DFS Traversal

Step 17: No new neighbors from J

Legend

Complete

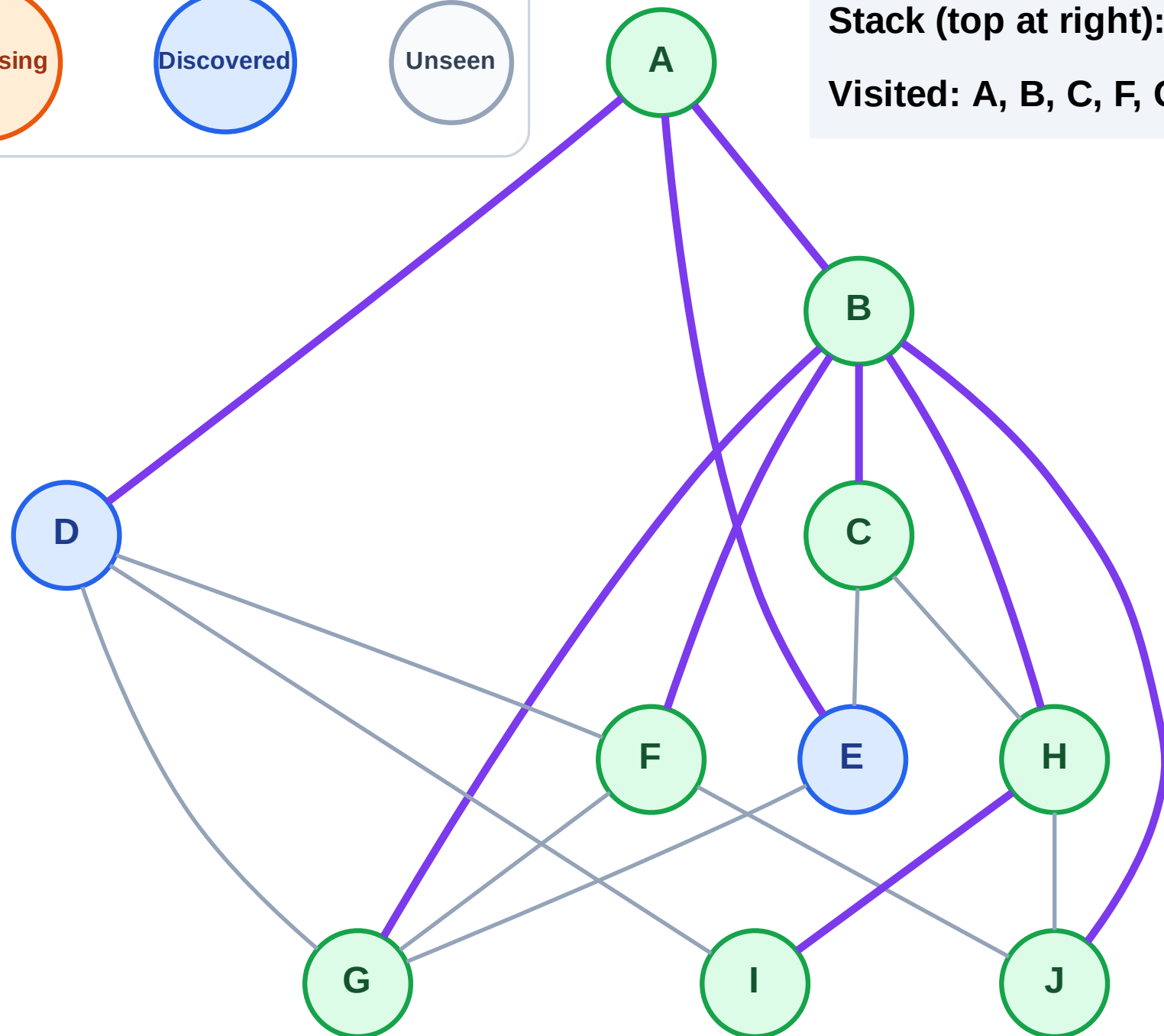
Processing

Discovered

Unseen

Stack (top at right): E | D

Visited: A, B, C, F, G, H, I, J



DFS Traversal

Step 18: Process node D

Legend

Complete

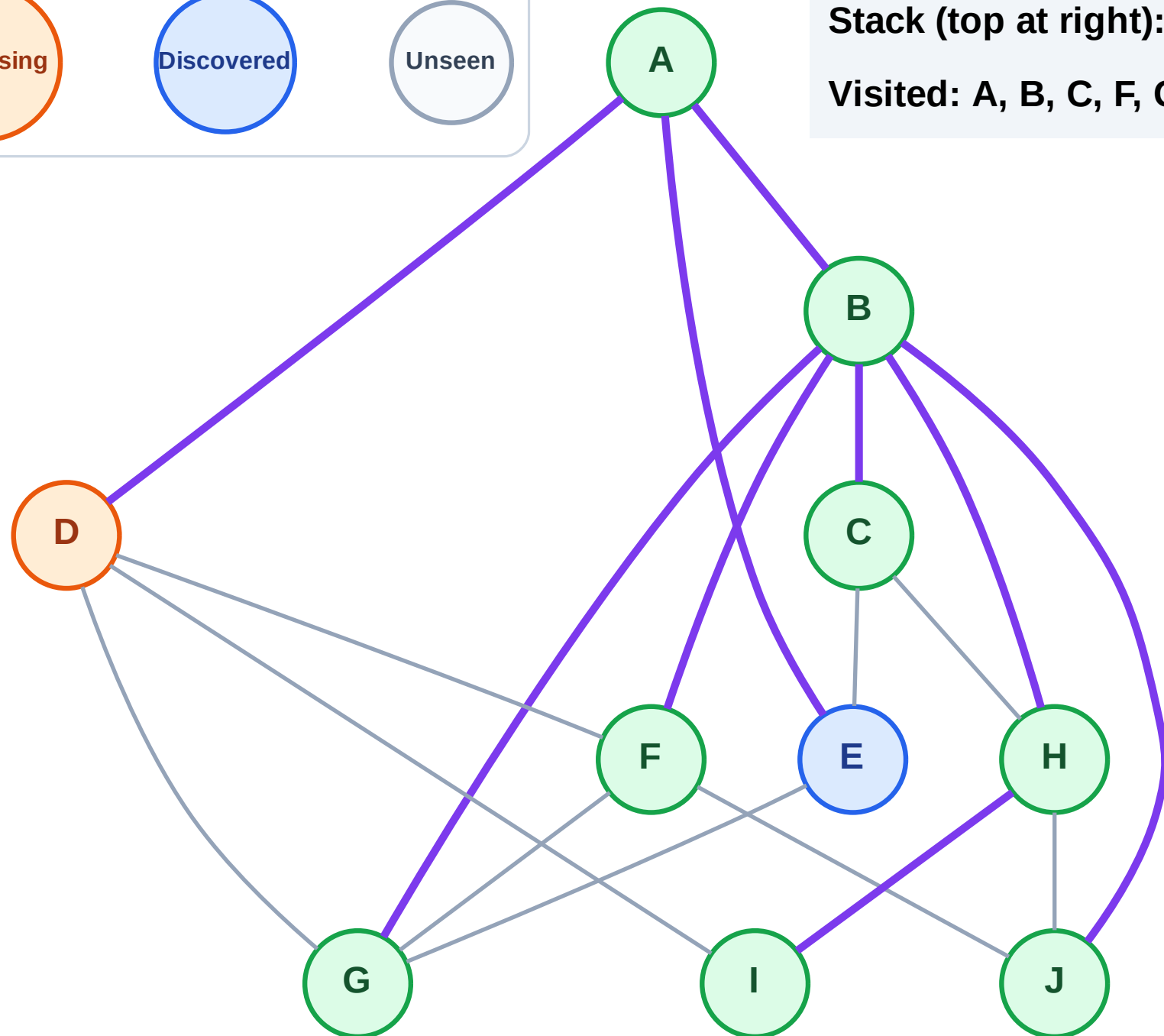
Processing

Discovered

Unseen

Stack (top at right): E

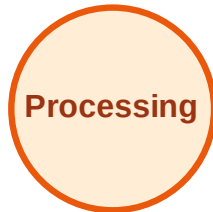
Visited: A, B, C, F, G, H, I, J



DFS Traversal

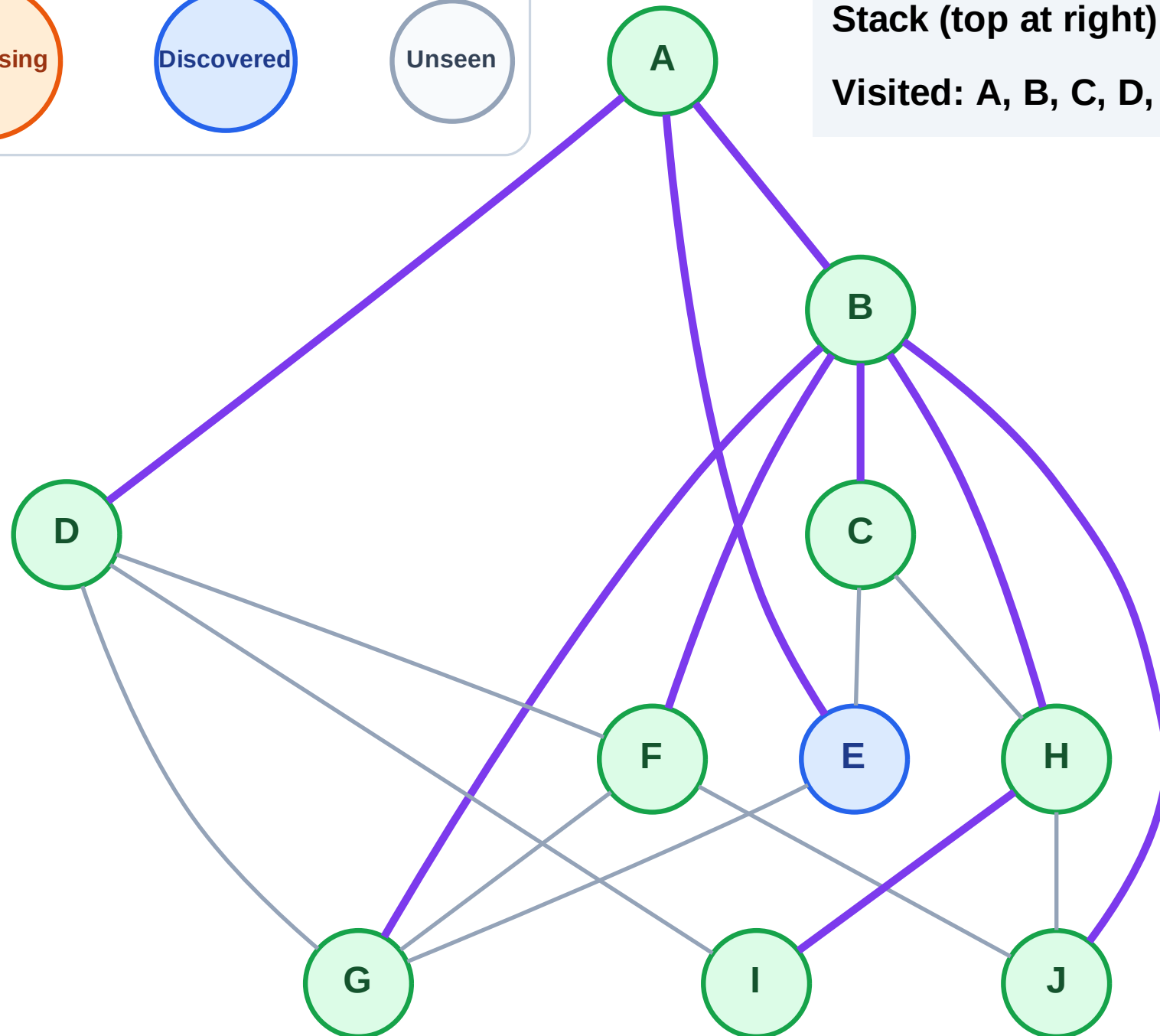
Step 19: No new neighbors from D

Legend



Stack (top at right): E

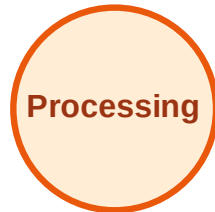
Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

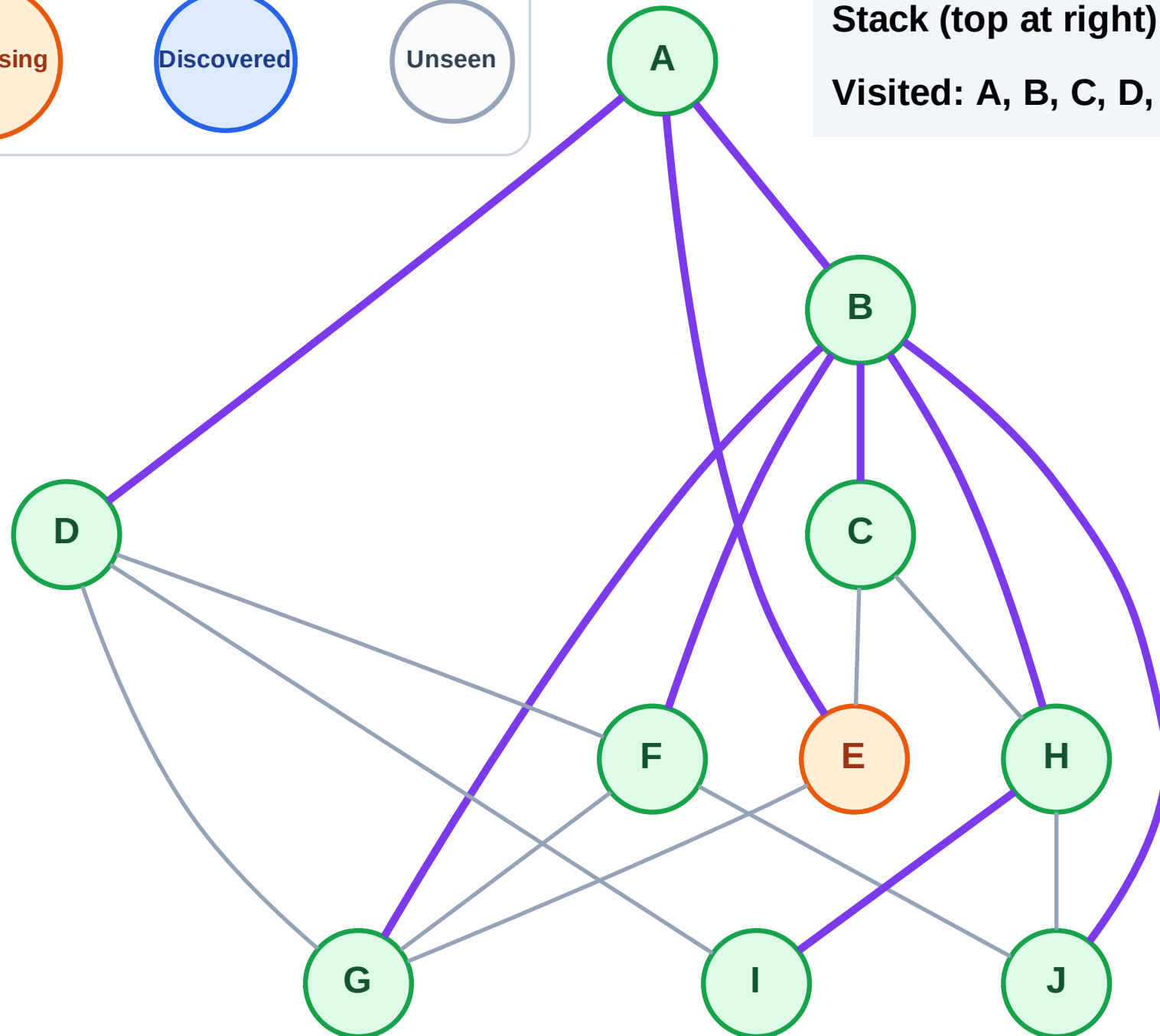
Step 20: Process node E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, F, G, H, I, J



DFS Traversal

Step 21: No new neighbors from E

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

